PT11: CONFIGURACIÓ D'ENTORNS SEGURS PER A L'EMMAGATZEMATGE DE DADES.

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Objectiu:

Aprendre a configurar un entron d'emmagatzematge de dades segur pels recursos compatits d'una organització implementant mètodes de protecció com RAID i còpies de seguretat. Apendre a límitar l'ùs de l'espai compartit per part d'usuaris i grups en un servidor de recursos compartits.

CONSIDERACIONS:

• Per a cada domini (midomXX.et i sisXX.et) cal fer un estudi de necessitats i explicar justificadament les següents eleccions:

◦ **Sistema RAID escollit**: cal escollir un tipus de RAID a implementar determinant el número de discos que el composaran, la seva finalitat i la mida d'aquests.

◦ **Planificació de BACKUPS:** cal escollir on s'emmagatzemaran les copies de seguretat i amb quina regularitat es realitzaran (definint els tipus de copia escollit en cada cas).

Es imporant valorar l'espai total necessàri segons la planificació escollida.

◦ **Configuració de quotes de disc:** cal determinar en quins discs, particions i/o directoris s'aplicaran límits a l'utilització de l'espai. Cal determinar quins límits s'aplicaran i de quin tipus seran.

Cal explicar si son els mateixos per a tots els usuaris o si variaran segons els grups.

• Amb les eleccions i justificacions clares, cal implementar les configuraciones a les Mvs disponibles afegint aquells elements d'emmagatzematge necessaris, instal·lant els rols i/o paquets necessaris i configurant les programacions de tasques necessàries.

• Amb l'ajuda dels equips clients dels dominis cal comprovar que les configuracions realitzades son funcionals i actuen tal com s'ha previst alhora implementar-les.

DOMINI 1: midom57.et (Windows Server)

Sistema RAID escollit:

RAID 5 (amb 4 discos de 1 TB cadascun)

- Total brut: 4 TB
- Espai útil aproximat: 3 TB
- Tolerància a fallada: 1 disc (Hot Spare)

Justificació: Aquest servidor allotja:

- Carpetes compartides d'usuaris
- Recursos departamentales
- Possiblement perfils mòbils

RAID 5 és un bon equilibri entre:

- Rendiment acceptable
- Bona capacitat útil
- Protección contra fallada d'un disc

Planificación de Backups

On s'emmagatzemen?

- Un **disco externo de 4 TB**
- O un NAS extern
- Mai al mateix RAID principal

Planificació proposada

Tipus còpia	Periodicitat	Retenció
Completa	Diumenge	4 setmanes
Incremental	Dilluns–Dissabte	2 setmanes

Quotes de disc

Implementació amb:

- **FSRM (File Server Resource Manager)**

¿Dónde aplicar quotes?

- Carpeta \\Servidor\Usuaris
- Carpeta \\Servidor\Departaments

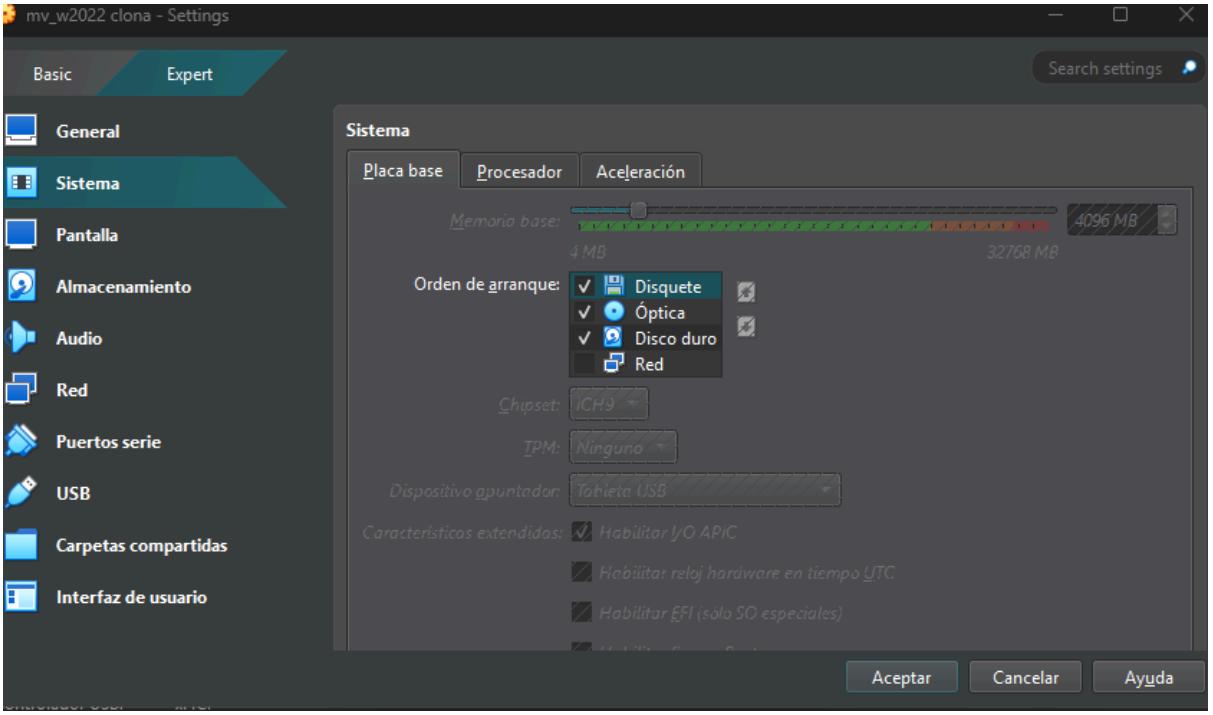
Límits proposats

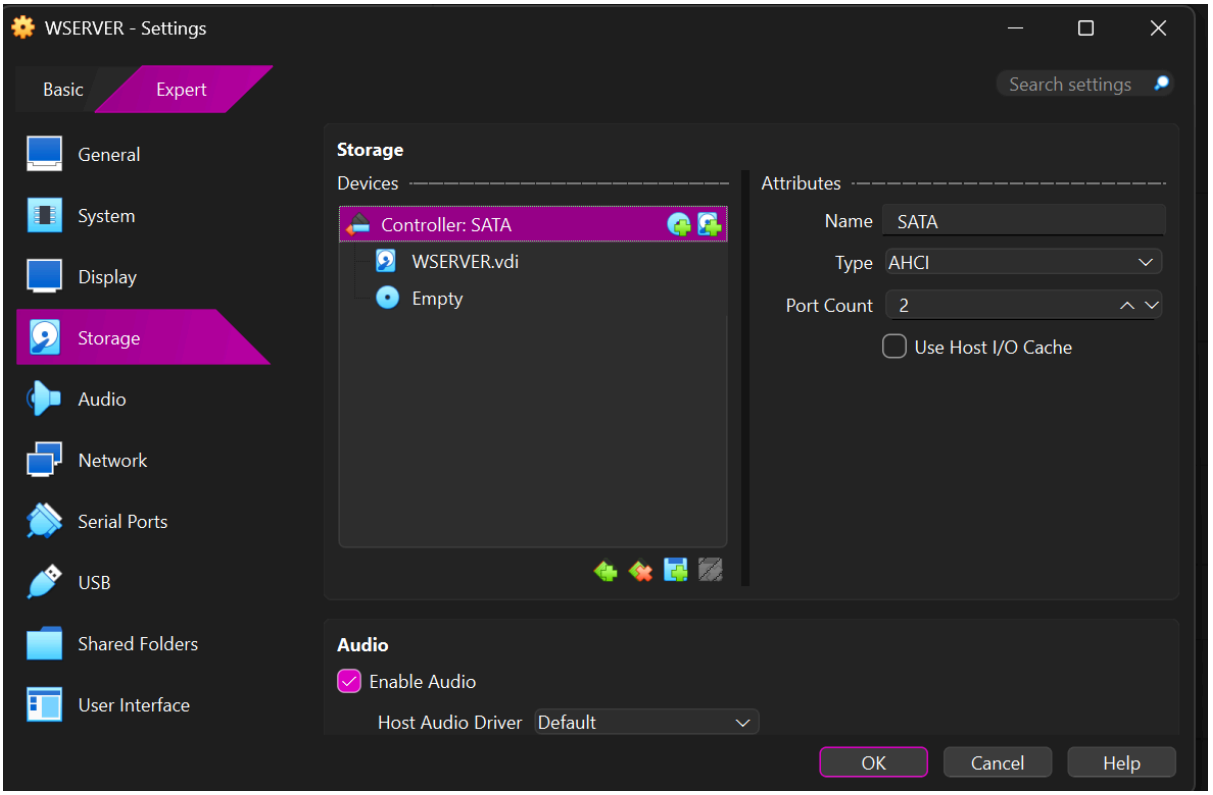
Grup	Límit	Tipus
Usuaris estàndard	10 GB	Hard quota
Admin	30 GB	Hard quota
Departaments	100 GB	Soft quota

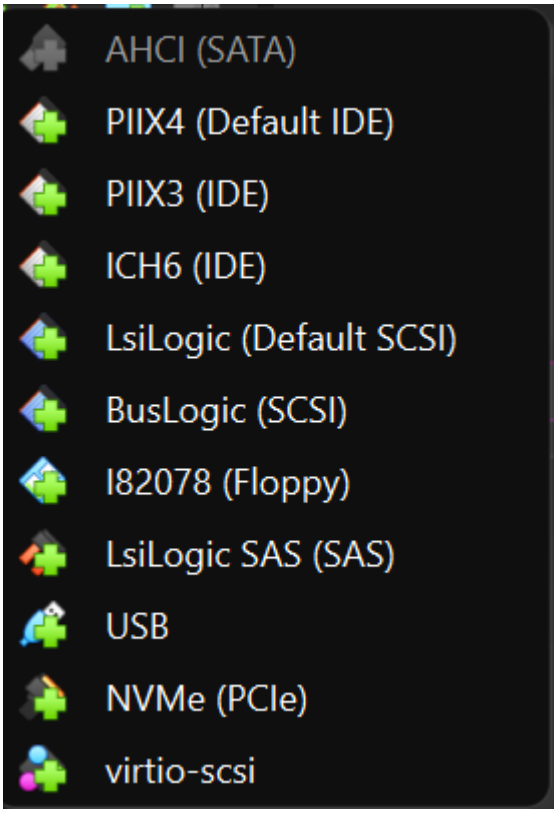
Justificació

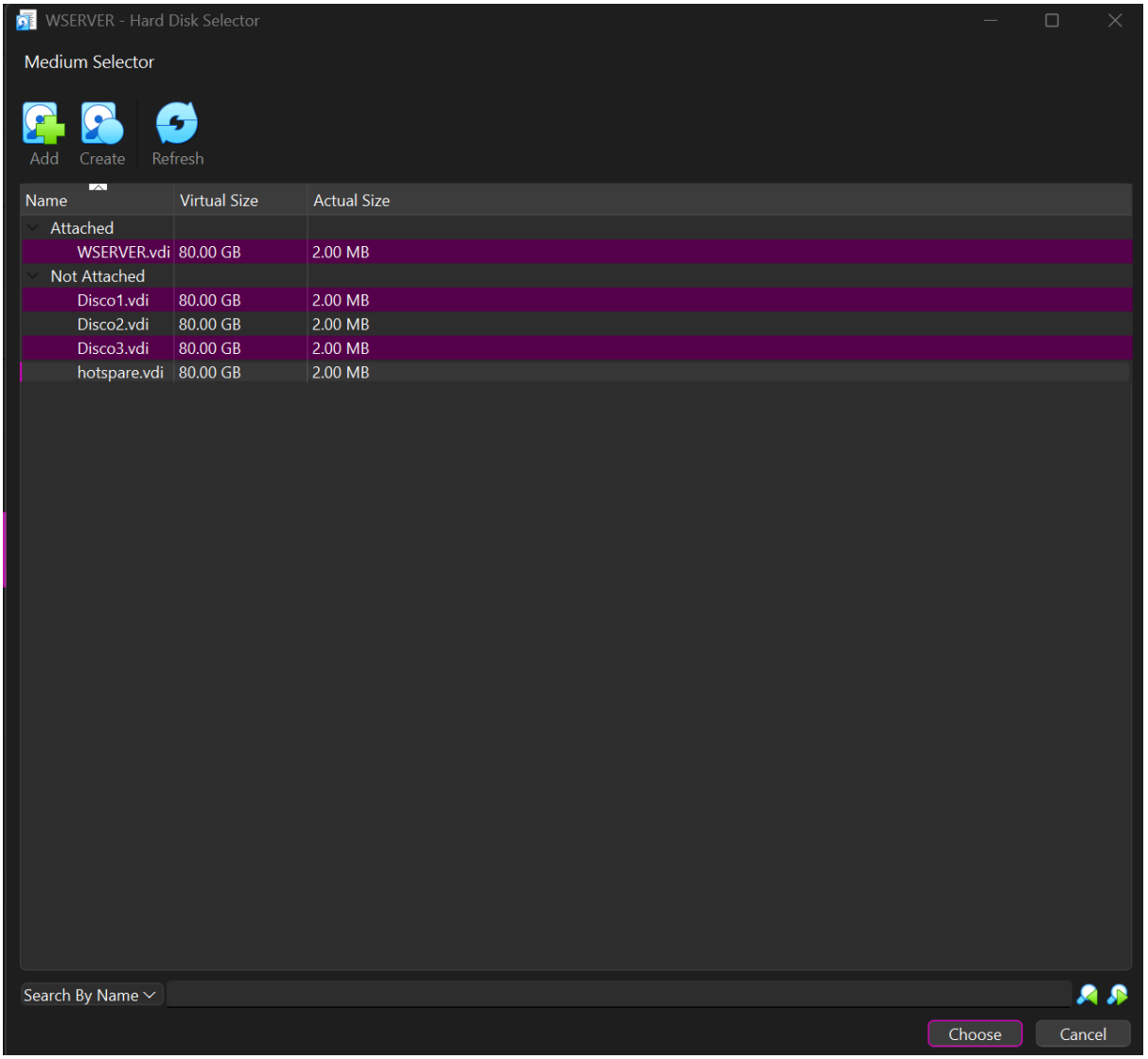
- Evita que un usuari saturi el sistema.
- Diferenciem segons responsabilitat.
- Hard quota bloqueja quan arriba al límit.
- Soft quota només avisa (ideal per departaments).

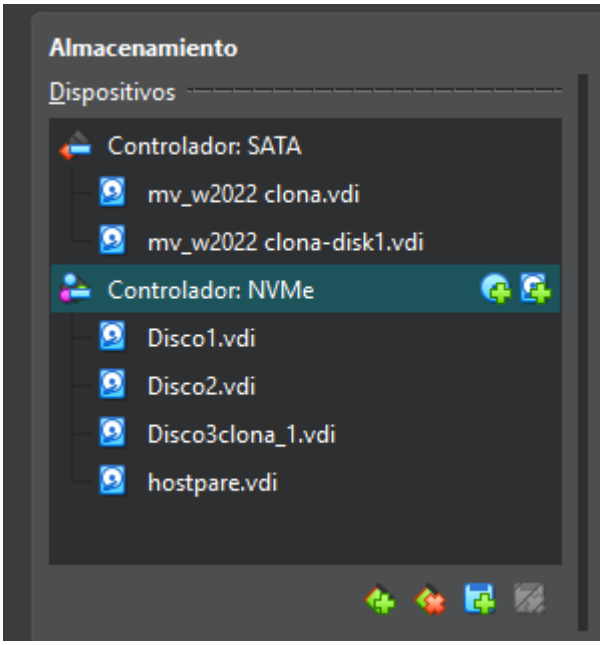
Antes de comenzar deberemos de cambiar la placa base de la máquina virtual a ICH9 debido a que nos dejará añadir 2 controladores sata y así nos vendrá mejor para la práctica.




WSERVER - Settings
Basic
Expert
Search settings
General
System
Display
Storage
Audio
Network
Serial Ports
USB
Shared Folders
User Interface
Storage
Devices
Controller: SATA
WSERVER.vdi
Empty
Attributes
Name
SATA
Type
AHCI
Port Count
2
Use Host I/O Cache
Audio
Enable Audio
Host Audio Driver
Default
OK
Cancel
Help


AHCI (SATA)
PIIX4 (Default IDE)
PIIX3 (IDE)
ICH6 (IDE)
LsiLogic (Default SCSI)
BusLogic (SCSI)
I82078 (Floppy)
LsiLogic SAS (SAS)
USB
NVMe (PCIe)
virtio-scsi

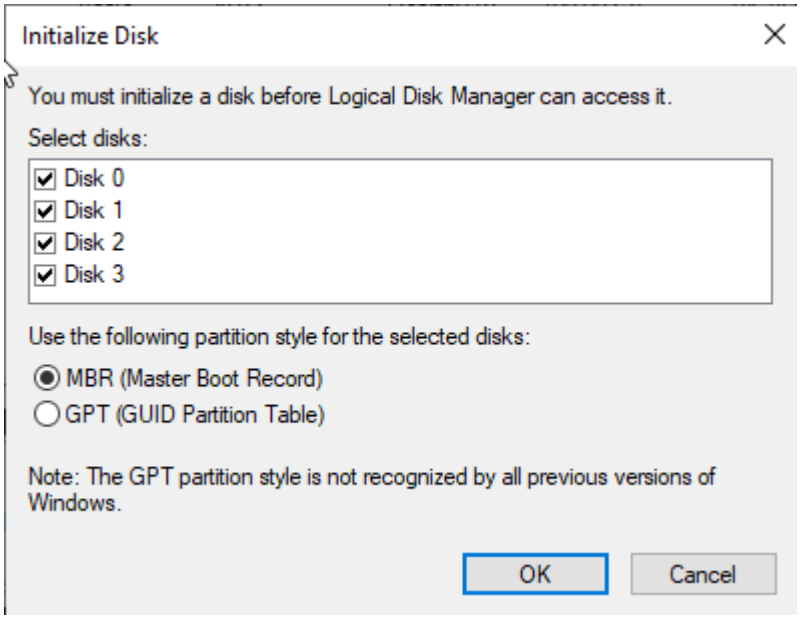

WSERVER - Hard Disk Selector
Medium Selector
Add
Create
Refresh
Name
Virtual Size
Actual Size
Attached
WSERVER.vdi 80.00 GB 2.00 MB
Not Attached
Disco1.vdi 80.00 GB 2.00 MB
Disco2.vdi 80.00 GB 2.00 MB
Disco3.vdi 80.00 GB 2.00 MB
hotspare.vdi 80.00 GB 2.00 MB
Search By Name
Choose
Cancel

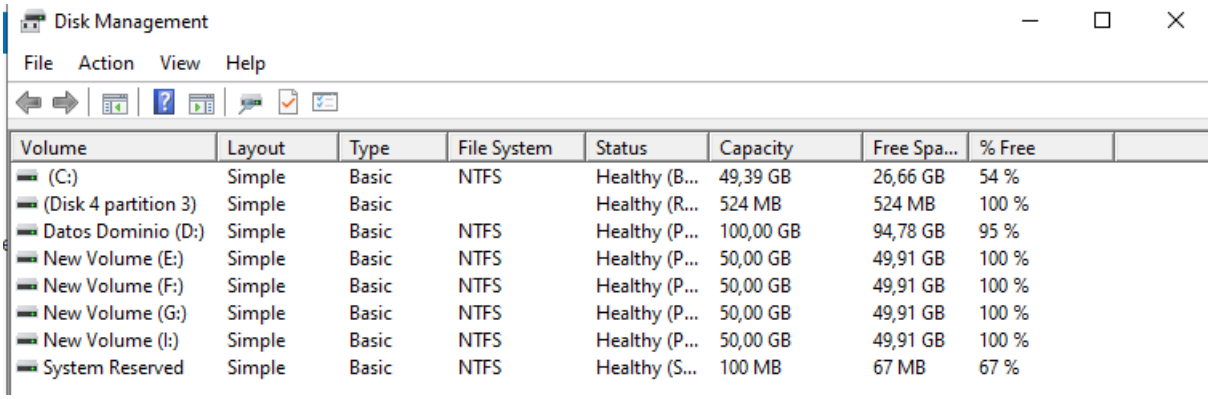

Almacenamiento
Dispositivos
Controlador: SATA
mv_w2022 clona.vdi
mv_w2022 clona-disk1.vdi
Controlador: NVMe
Disco1.vdi
Disco2.vdi
Disco3clona_1.vdi
hostpare.vdi

RAID con Storage Spaces

Paso previo

Antes de comenzar deberemos de inicializar los discos y darles el formato correcto para su utilización para llevar a cabo esto deberemos de irnos al administrador de discos de sistema.





Volume	Layout	Type	File System	Status	Capacity	Free Spa...	% Free
(C:)	Simple	Basic	NTFS	Healthy (B...	49,39 GB	26,66 GB	54 %
(Disk 4 partition 3)	Simple	Basic		Healthy (R...	524 MB	524 MB	100 %
Datos Dominio (D:)	Simple	Basic	NTFS	Healthy (P...	100,00 GB	94,78 GB	95 %
New Volume (E:)	Simple	Basic	NTFS	Healthy (P...	50,00 GB	49,91 GB	100 %
New Volume (F:)	Simple	Basic	NTFS	Healthy (P...	50,00 GB	49,91 GB	100 %
New Volume (G:)	Simple	Basic	NTFS	Healthy (P...	50,00 GB	49,91 GB	100 %
New Volume (I:)	Simple	Basic	NTFS	Healthy (P...	50,00 GB	49,91 GB	100 %
System Reserved	Simple	Basic	NTFS	Healthy (S...	100 MB	67 MB	67 %

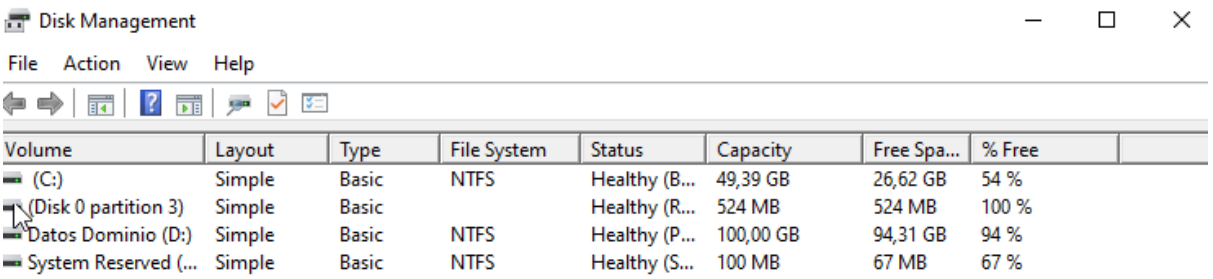
Una vez inicializados deberemos de crear el Raid desde PowerShell.

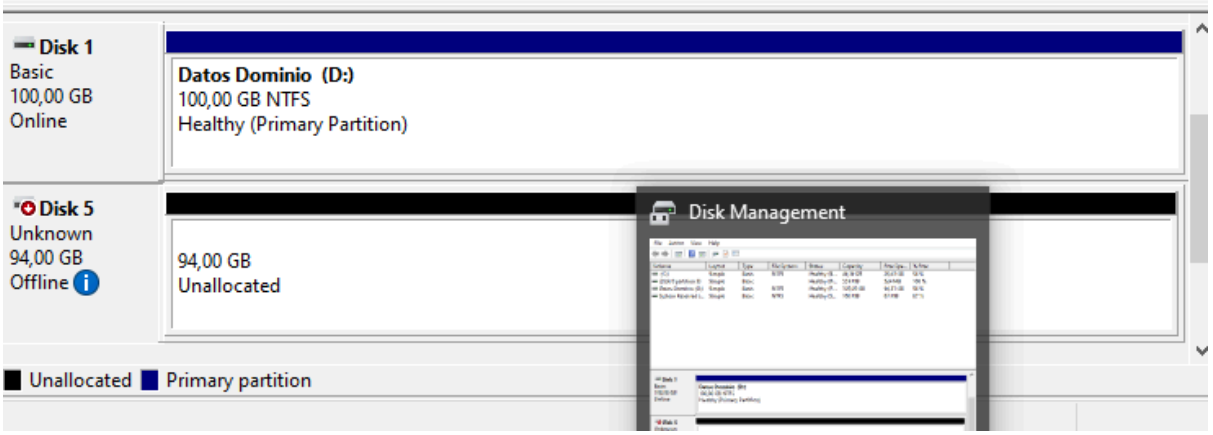
2 Crea el Storage Pool

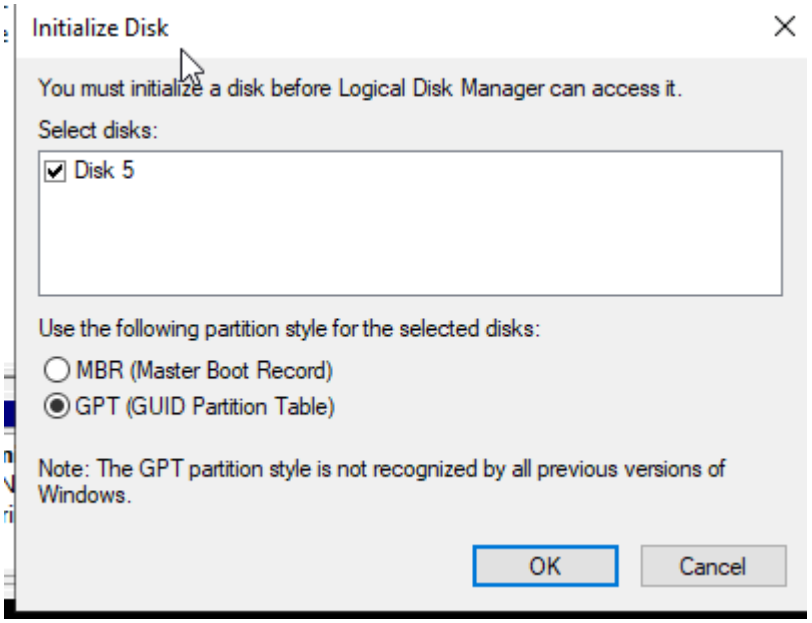
```
PS C:\Users\Administrator> New-StoragePool -FriendlyName "MiPoolRaid5" -StorageSubSystemFriendlyName "Windows Storage*" -Phys
icalDisks (Get-PhysicalDisk | Where-Object CanPool -eq $True)

FriendlyName OperationalStatus HealthStatus IsPrimordial IsReadOnly      Size AllocatedSize
------------ ----------------- ------------ ------------ ----------      ---- -------------
MiPoolRaid5  OK                Healthy      False        False      148.46 GB        768 MB


PS C:\Users\Administrator>
```

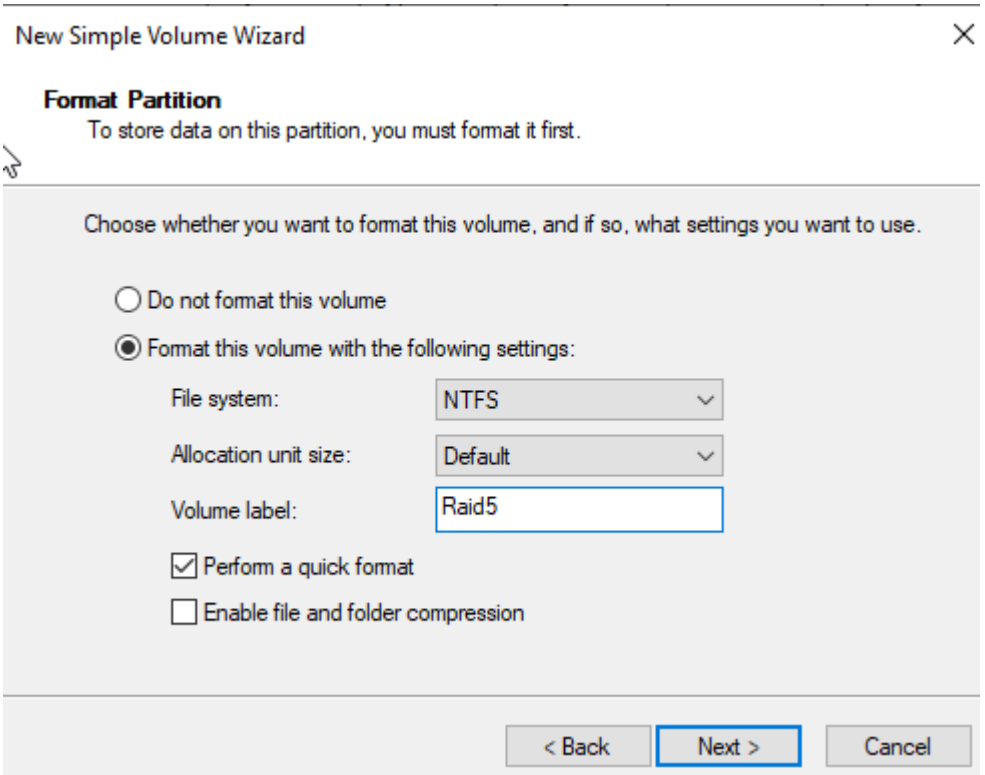
3 Crea el volumen con resiliencia de paridad (RAID 5)

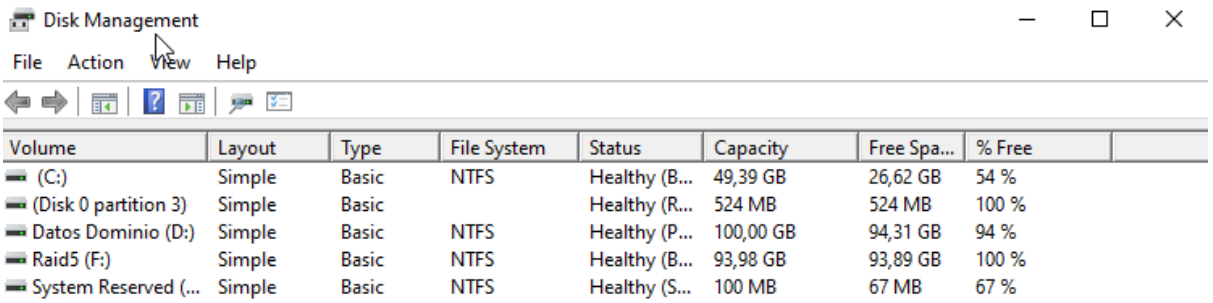
```
PS C:\Users\Administrator> New-VirtualDisk -StoragePoolFriendlyName "MiPoolRaid5" -FriendlyName "Raid5_Volume" -ProvisioningT
ype Fixed -ResiliencySettingName Parity -UseMaximumSize

FriendlyName ResiliencySettingName FaultDomainRedundancy OperationalStatus HealthStatus  Size FootprintOnPool StorageEfficie
                                                                                                                      ncy
------------ --------------------- --------------------- ----------------- ------------  ---- --------------- -------------
Raid5_Volume Parity                1                     OK                Healthy      94 GB        144.5 GB        65,05 %


PS C:\Users\Administrator> _
```



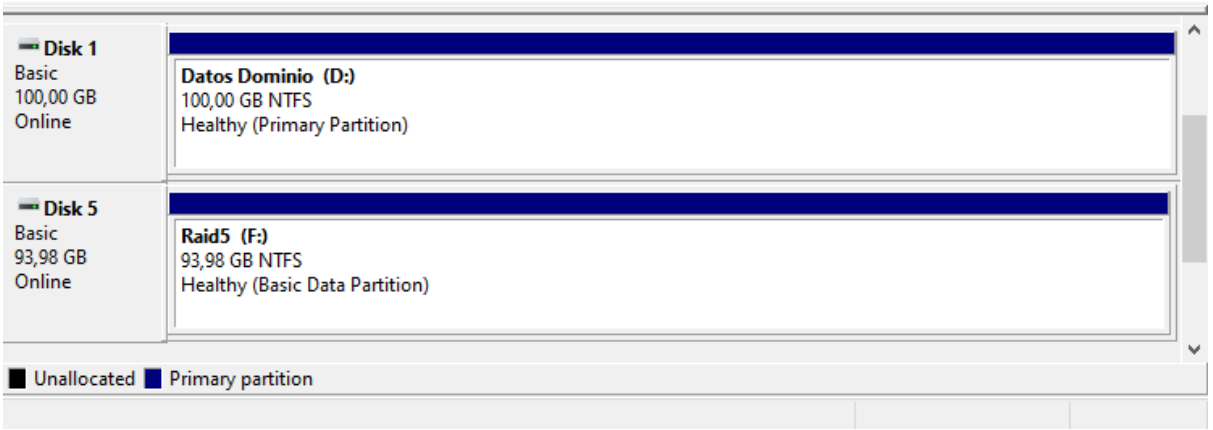


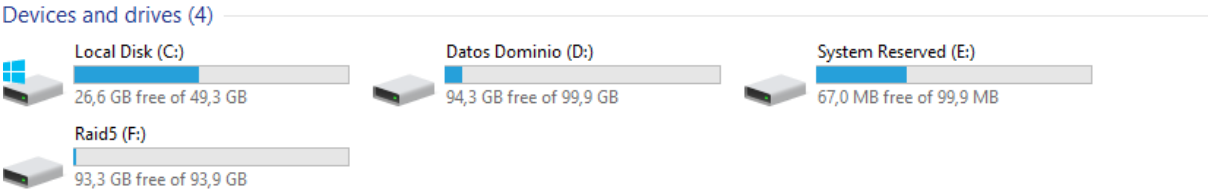

Initialize Disk
You must initialize a disk before Logical Disk Manager can access it.
Select disks:
Disk 5
Use the following partition style for the selected disks:
MBR (Master Boot Record)
GPT (GUID Partition Table)
Note: The GPT partition style is not recognized by all previous versions of Windows.
OK
Cancel


New Simple Volume Wizard
Format Partition
To store data on this partition, you must format it first.
Choose whether you want to format this volume, and if so, what settings you want to use.
Do not format this volume
Format this volume with the following settings:
File system:
NTFS
Allocation unit size:
Default
Volume label:
Raid5
Perform a quick format
Enable file and folder compression
< Back
Next >
Cancel

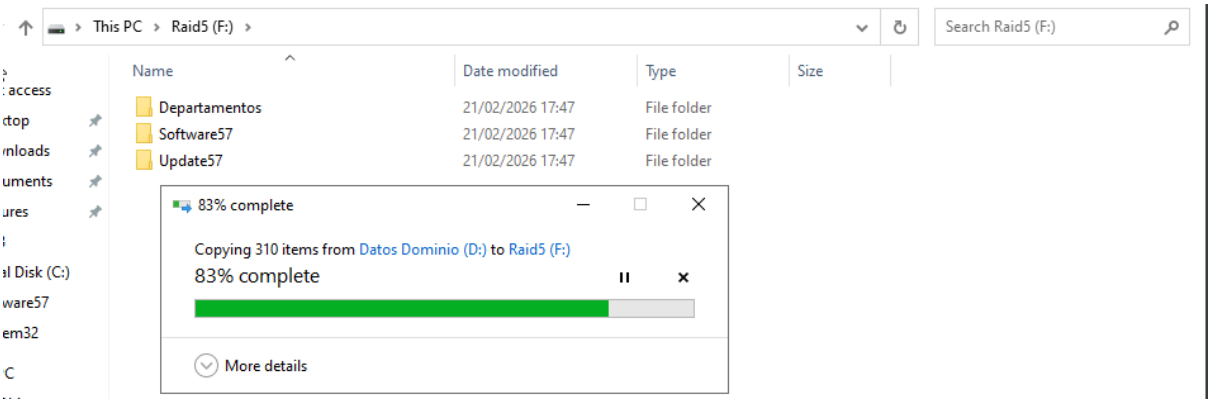


Volume	Layout	Type	File System	Status	Capacity	Free Spa...	% Free
(C:)	Simple	Basic	NTFS	Healthy (B...	49,39 GB	26,62 GB	54 %
(Disk 0 partition 3)	Simple	Basic		Healthy (R...	524 MB	524 MB	100 %
Datos Dominio (D:)	Simple	Basic	NTFS	Healthy (P...	100,00 GB	94,31 GB	94 %
Raid5 (F:)	Simple	Basic	NTFS	Healthy (B...	93,98 GB	93,89 GB	100 %
System Reserved (...	Simple	Basic	NTFS	Healthy (S...	100 MB	67 MB	67 %

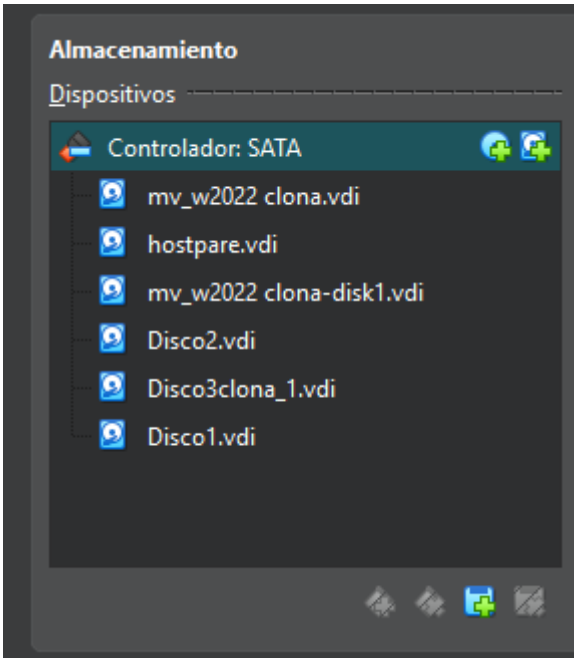




Movemos los archivos al Raid



Añadimos el hotspare a la lista de discos de la máquina virtual.



1 Añádelo al pool

```
PS C:\Users\Administrator> Add-PhysicalDisk -StoragePoolFriendlyName "MiPoolRaid5" -PhysicalDisks (Get-PhysicalDisk | Where-
bject CanPool -eq $True)
PS C:\Users\Administrator>
```

2 Márcalo como Hot Spare

```
PS C:\Users\Administrator> $disk = Get-PhysicalDisk | Where-Object SerialNumber -eq "VBf6fe065a-3ccb21f5"
PS C:\Users\Administrator> Set-PhysicalDisk -InputObject $disk -Usage HotSpare
```

(Lo he tenido que hacer de esta forma debido a que la he liado antes y ahora no podía hacerlo y era la única solución que he encontrado)

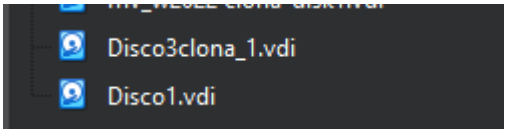
Comprobación:

```
PS C:\Users\Administrator> Get-StoragePool | Select FriendlyName, HealthStatus, OperationalStatus

FriendlyName HealthStatus OperationalStatus
------------ ------------ -----------------
Primordial   Healthy      OK
MiPoolRaid5  Healthy      OK


PS C:\Users\Administrator>
```

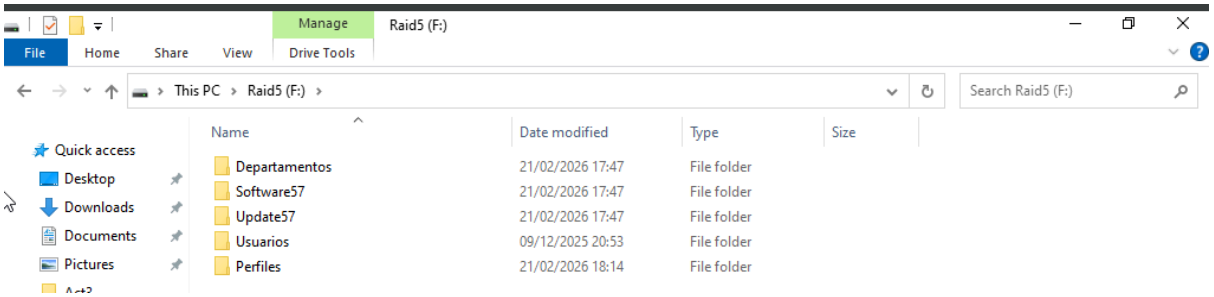
```
PS C:\Users\Administrator> Get-VirtualDisk | Select FriendlyName, ResiliencySettingName, HealthStatus, OperationalStatus

FriendlyName ResiliencySettingName HealthStatus OperationalStatus
------------ --------------------- ------------ -----------------
Raid5_Volume Parity                Healthy      OK


PS C:\Users\Administrator>
```

```
PS C:\Users\Administrator> Get-PhysicalDisk | Format-Table FriendlyName, SerialNumber, Usage, HealthStatus -AutoSize

FriendlyName  SerialNumber        Usage       HealthStatus
------------  ------------        -----       ------------
VBOX HARDDISK VB42738b08-9566193e Auto-Select Healthy
VBOX HARDDISK VB1bba537f-9db6af5c Auto-Select Healthy
VBOX HARDDISK VB5619da6e-94657a96 Auto-Select Healthy
VBOX HARDDISK VB8f1c80b9-4b2abeb1 Auto-Select Healthy
VBOX HARDDISK VBf77fd448-b60e6355 Auto-Select Healthy
VBOX HARDDISK VBf6fe065a-3ccb21f5 Hot Spare   Healthy


PS C:\Users\Administrator>
```

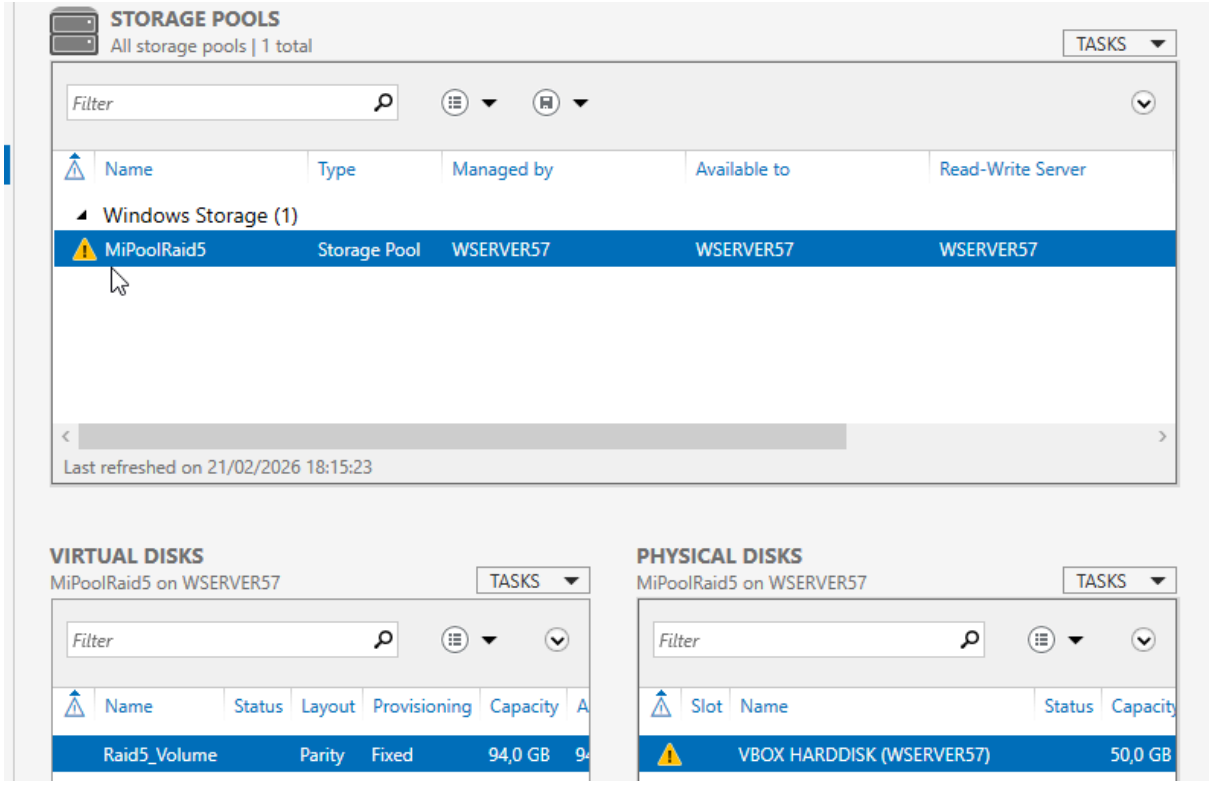
Quitamos el disco numero 2



Seguimos teniendo las carpetas y todo dentro del raid



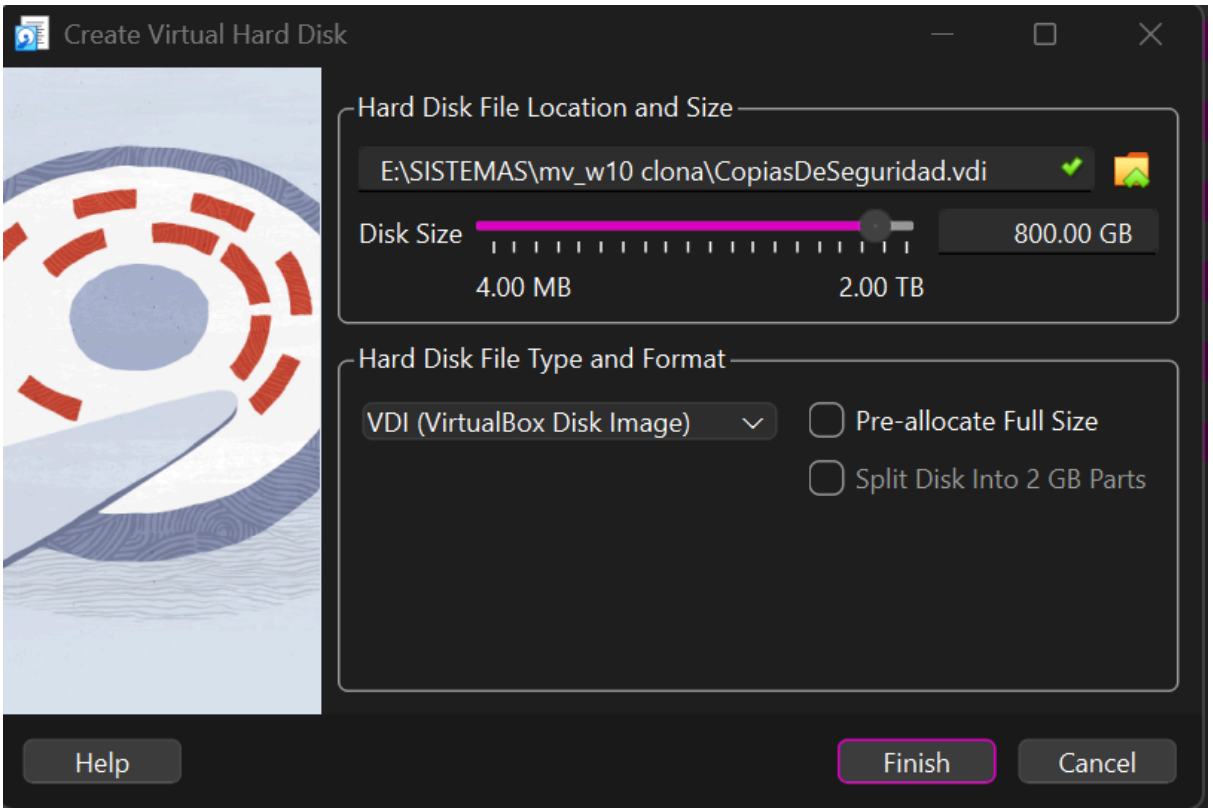
Nos saltara una advertencia a continuación:



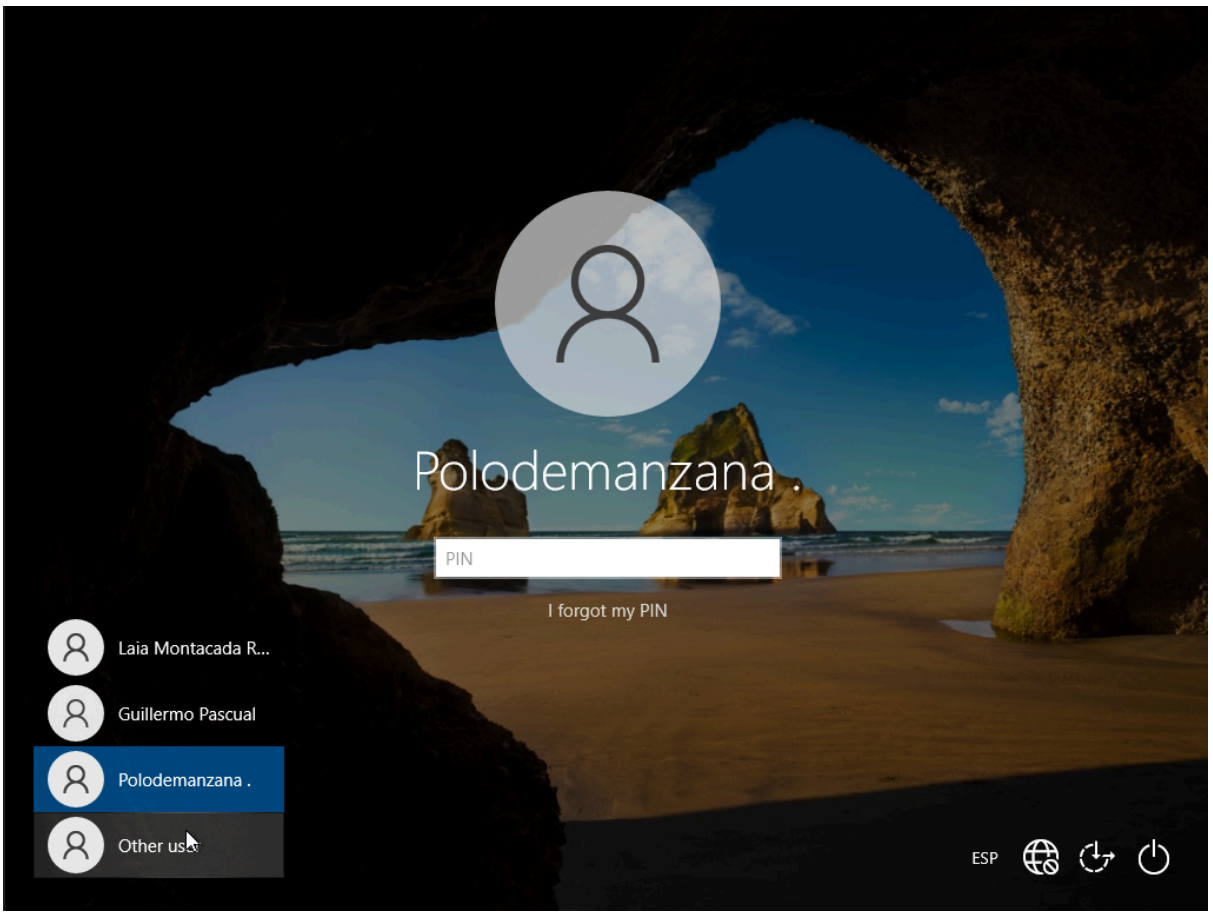
BackUPs

Windows Server 2022

Teniendo en cuenta de que el cambio diario de almacenamiento sería de un 5% de la copia total de 1,5TB nos daría una cantidad de aumento diario de 75GB , lo que nos daría semanalmente 1.95TB y mensualmente redondeando a 8TB lo cual utilizaremos un 10% para la práctica 800GB.



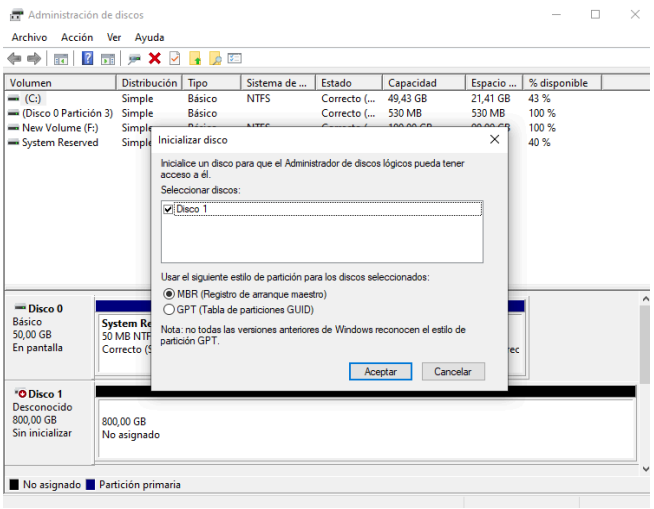
Deberemos iniciar sesión con el administrador local del dispositivo en el cual se encontrara el disco de los backups.



🧱 PASO 1 — Preparar el disco de backup

Conecta el disco externo (por ejemplo 8TB).

Ve a:

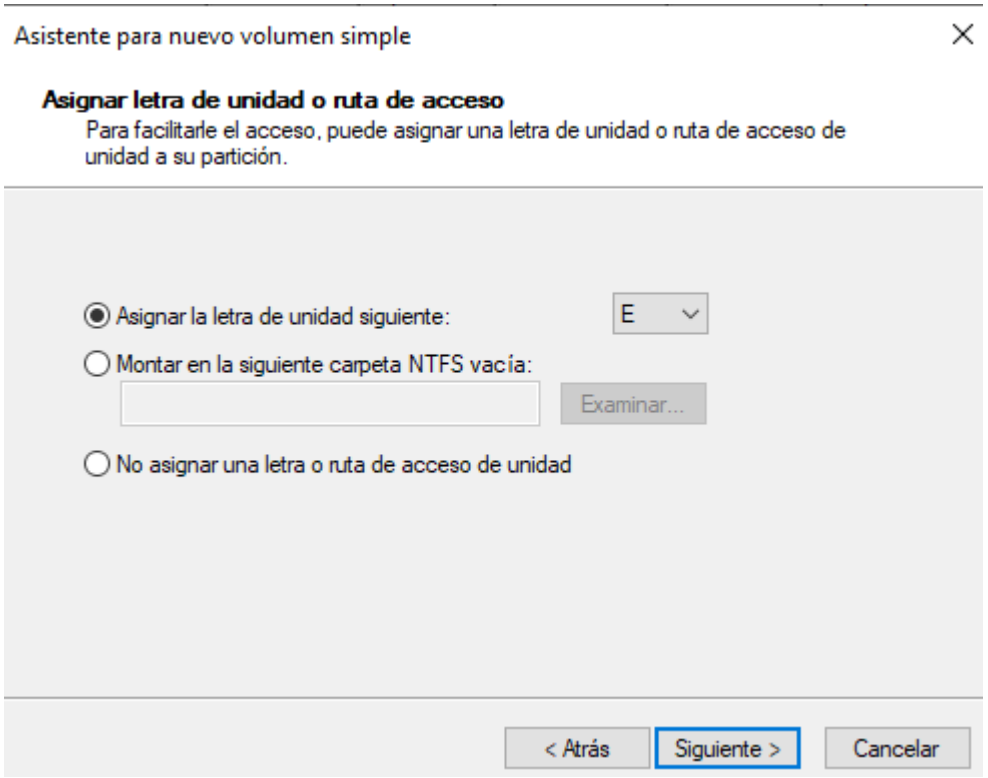


Crea un nuevo volumen simple.

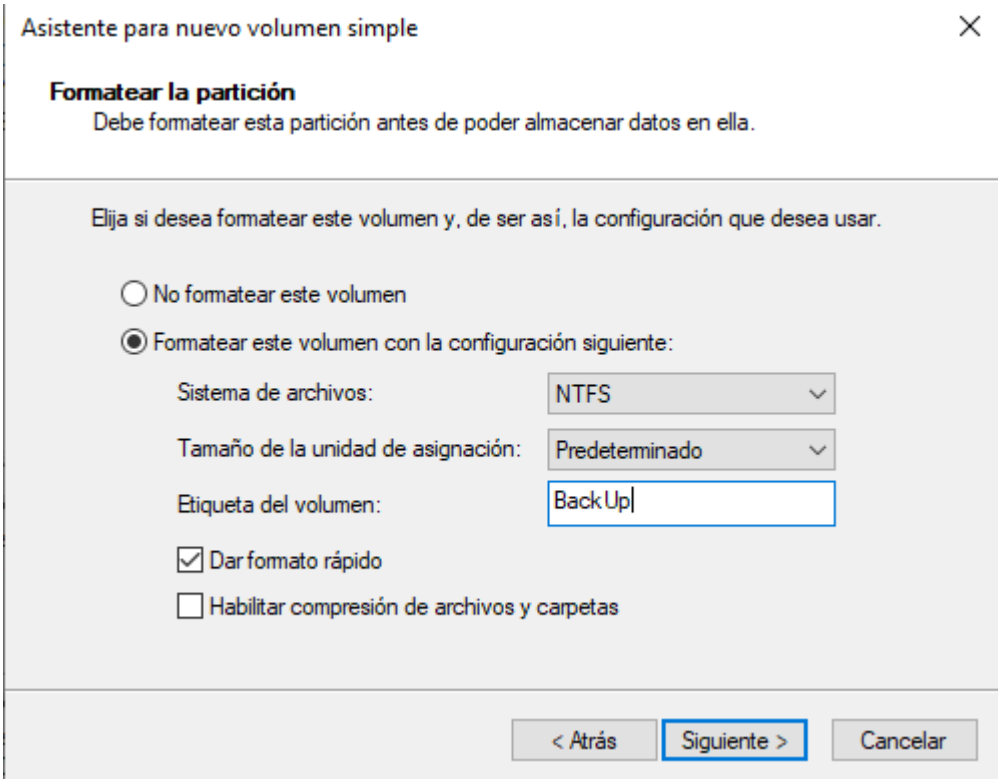
Formatea en:

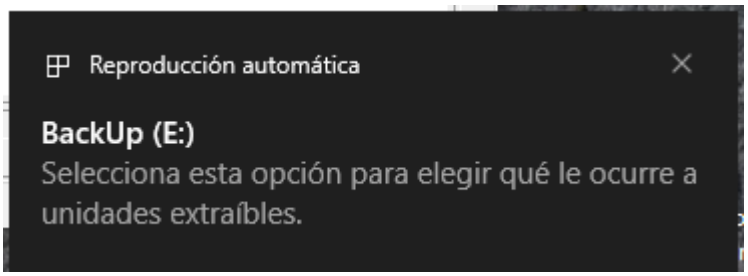
- **NTFS**

Asigna una letra (ejemplo: `E:`)



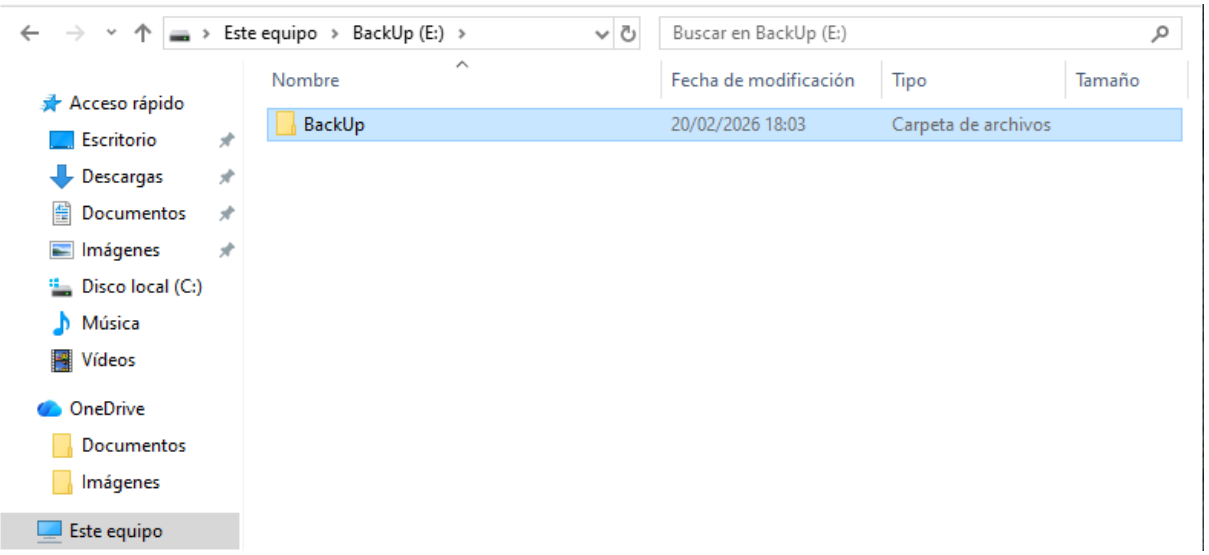
Nómbralo: `BACKUP`

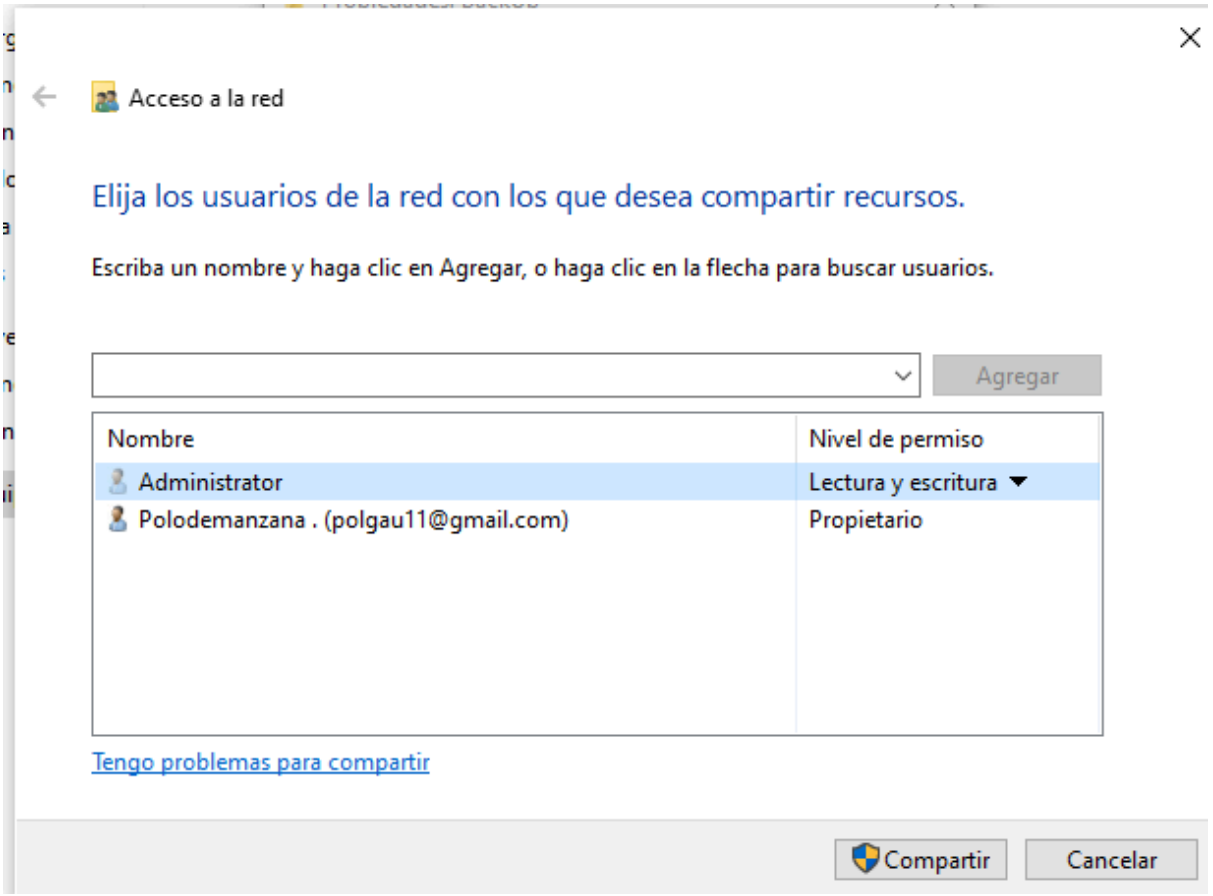


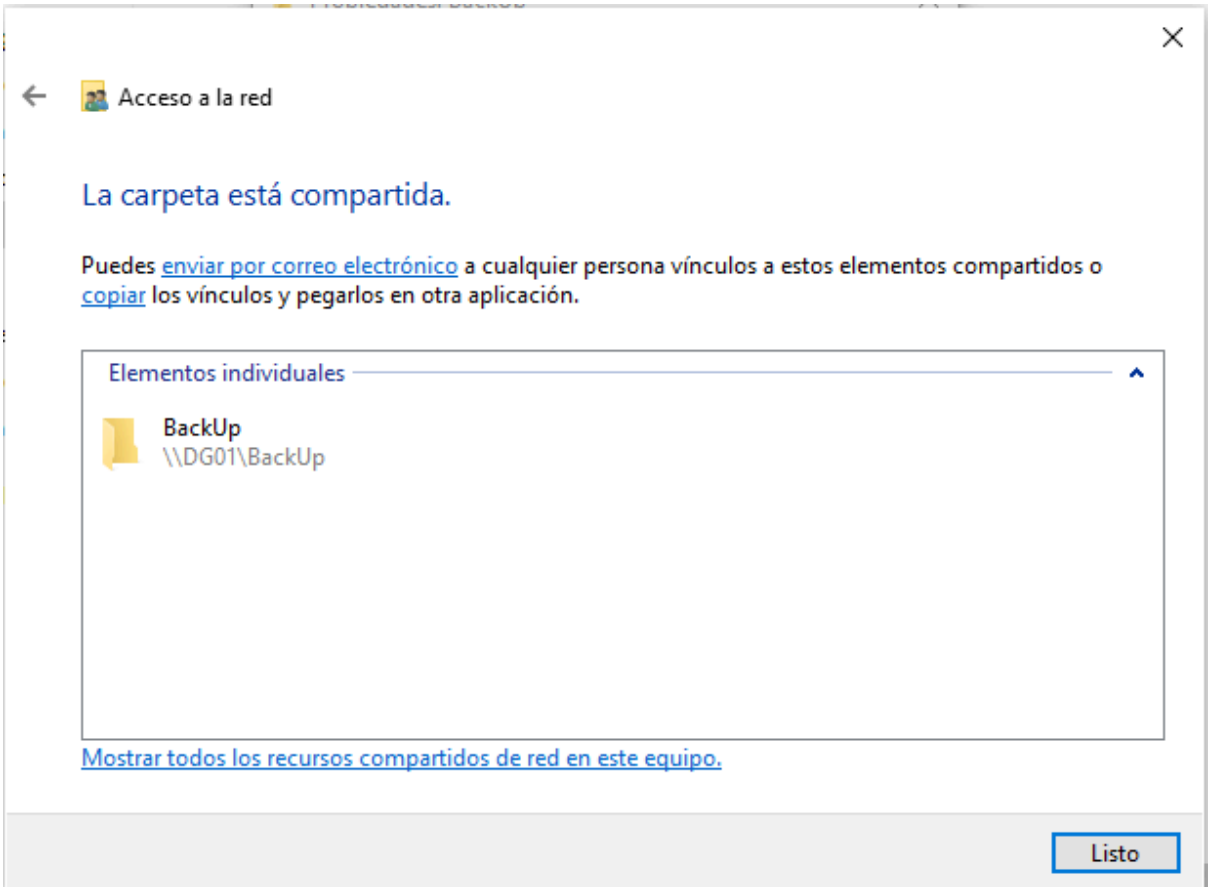


🗄PASO 2 - Compartir carpeta BackUp

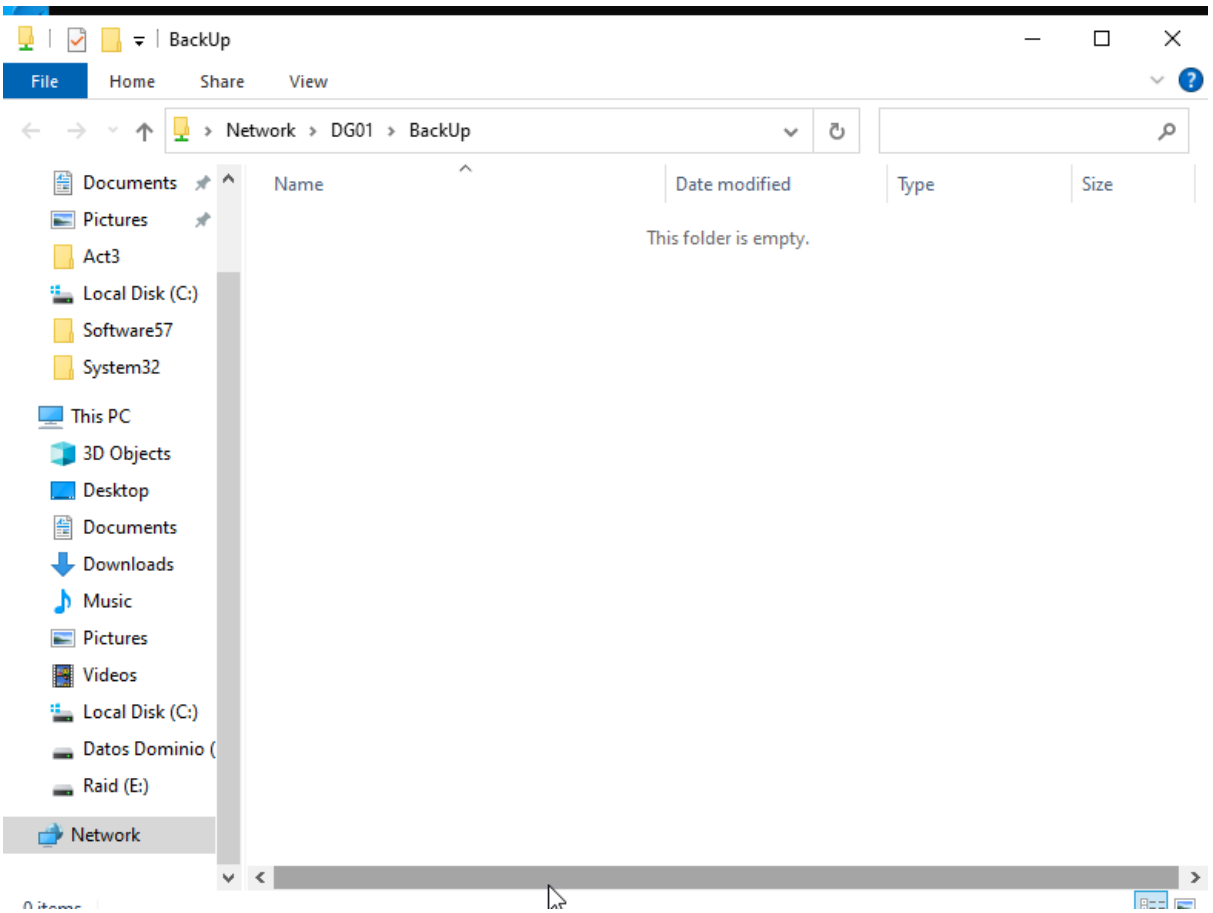
Deberemos crear una carpeta donde se guardarán los BackUp hechos dentro del del disco que hemos formateado y compartirlo para obtener la ruta UNC.


Este equipo › BackUp (E:) ›
Buscar en BackUp (E:)
Acceso rápido
Escritorio
Descargas
Documentos
Imágenes
Disco local (C:)
Música
Vídeos
OneDrive
Documentos
Imágenes
Este equipo
Nombre
Fecha de modificación
Tipo
Tamaño
BackUp
20/02/2026 18:03
Carpeta de archivos


Acceso a la red
Elija los usuarios de la red con los que desea compartir recursos.
Escriba un nombre y haga clic en Agregar, o haga clic en la flecha para buscar usuarios.
Agregar
Nombre
Nivel de permiso
Administrator
Lectura y escritura
Polodemanzana . (polgau11@gmail.com)
Propietario
Tengo problemas para compartir
Compartir
Cancelar



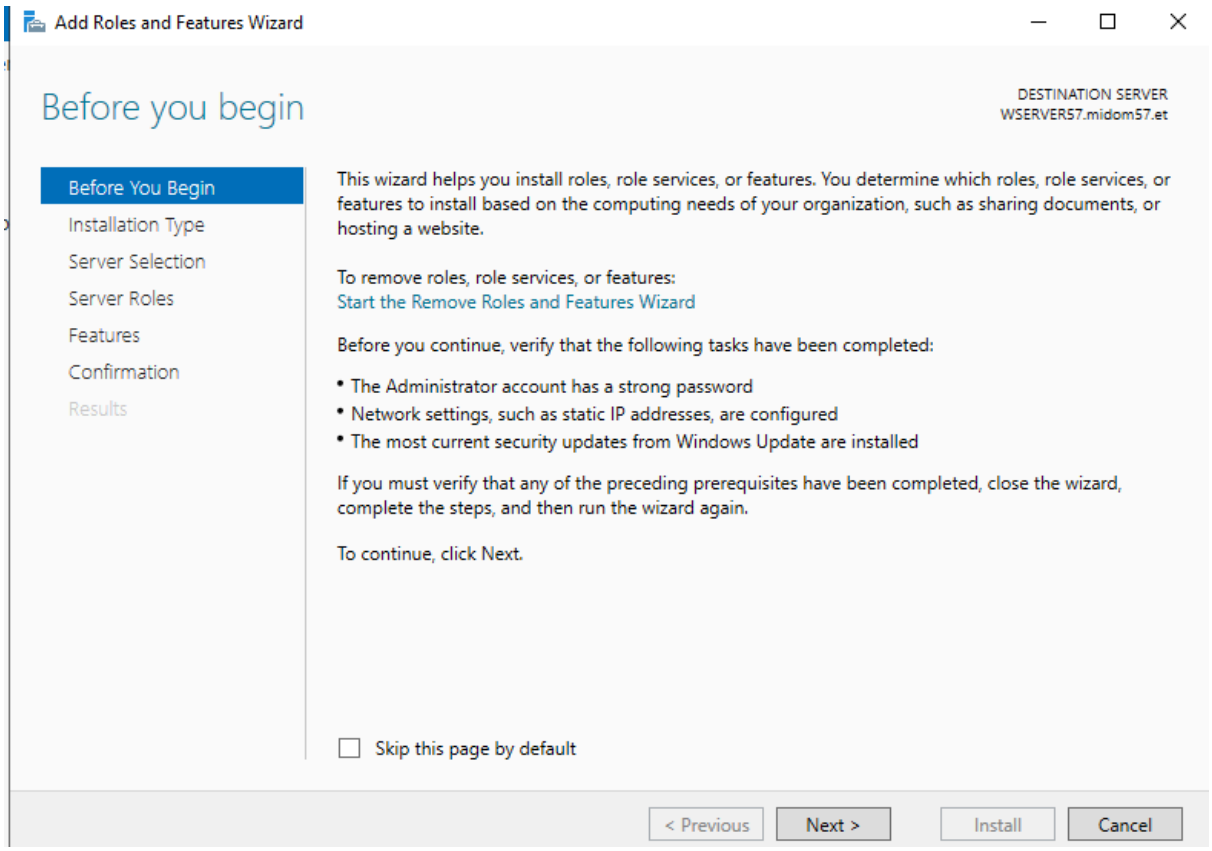
Una vez la hemos compartido revisamos en el servidor que tengamos acceso a la carpeta:



🛠️ PASO 3 — Instalar Windows Server Backup

Ve a:

Administrador del servidor → Agregar roles y características

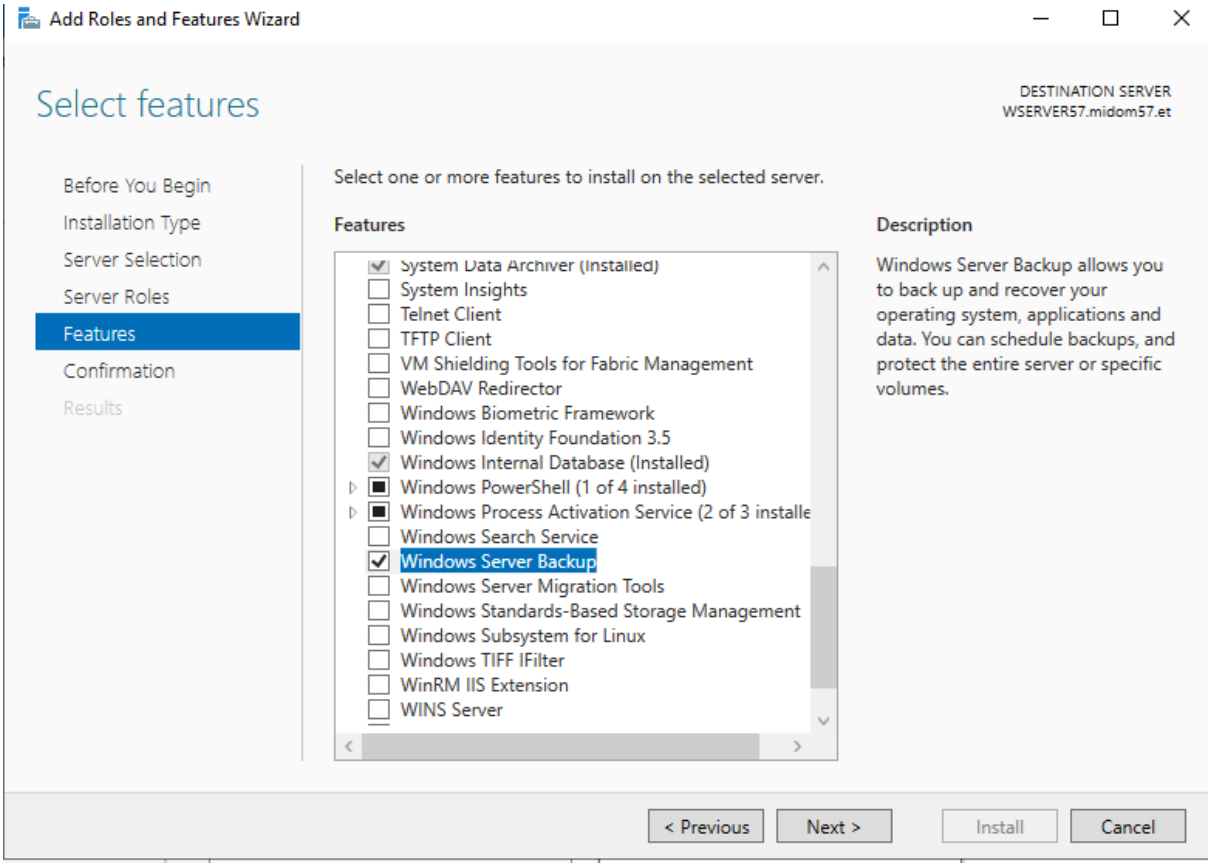


Tipo de instalación:

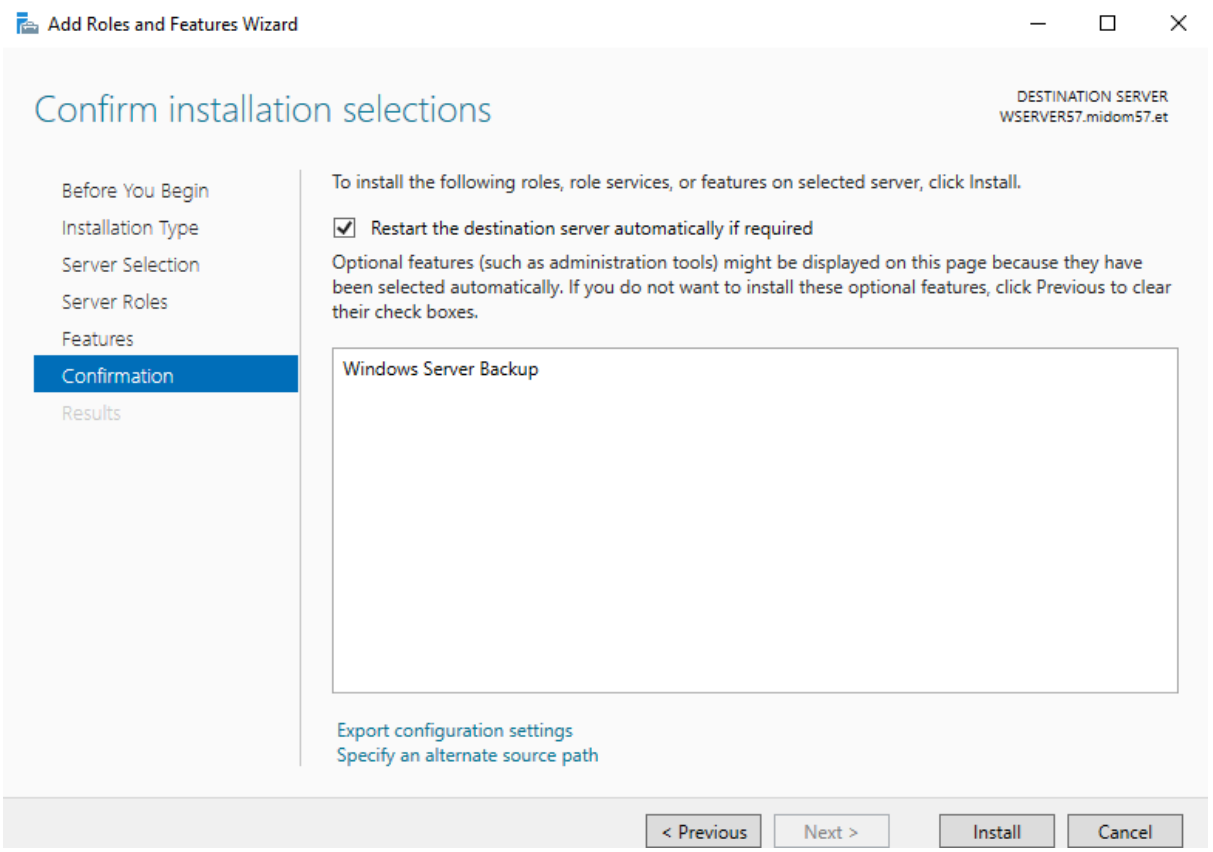
Basada en características

Marca:

Windows Server Backup


Add Roles and Features Wizard
Select features
DESTINATION SERVER
WSERVER57.midom57.et
Before You Begin
Installation Type
Server Selection
Server Roles
Features
Confirmation
Results
Select one or more features to install on the selected server.
Features
System Data Archiver (Installed)
System Insights
Telnet Client
TFTP Client
VM Shielding Tools for Fabric Management
WebDAV Redirector
Windows Biometric Framework
Windows Identity Foundation 3.5
Windows Internal Database (Installed)
Windows PowerShell (1 of 4 installed)
Windows Process Activation Service (2 of 3 installe
Windows Search Service
Windows Server Backup
Windows Server Migration Tools
Windows Standards-Based Storage Management
Windows Subsystem for Linux
Windows TIFF IFilter
WinRM IIS Extension
WINS Server
Description
Windows Server Backup allows you to back up and recover your operating system, applications and data. You can schedule backups, and protect the entire server or specific volumes.
< Previous
Next >
Install
Cancel

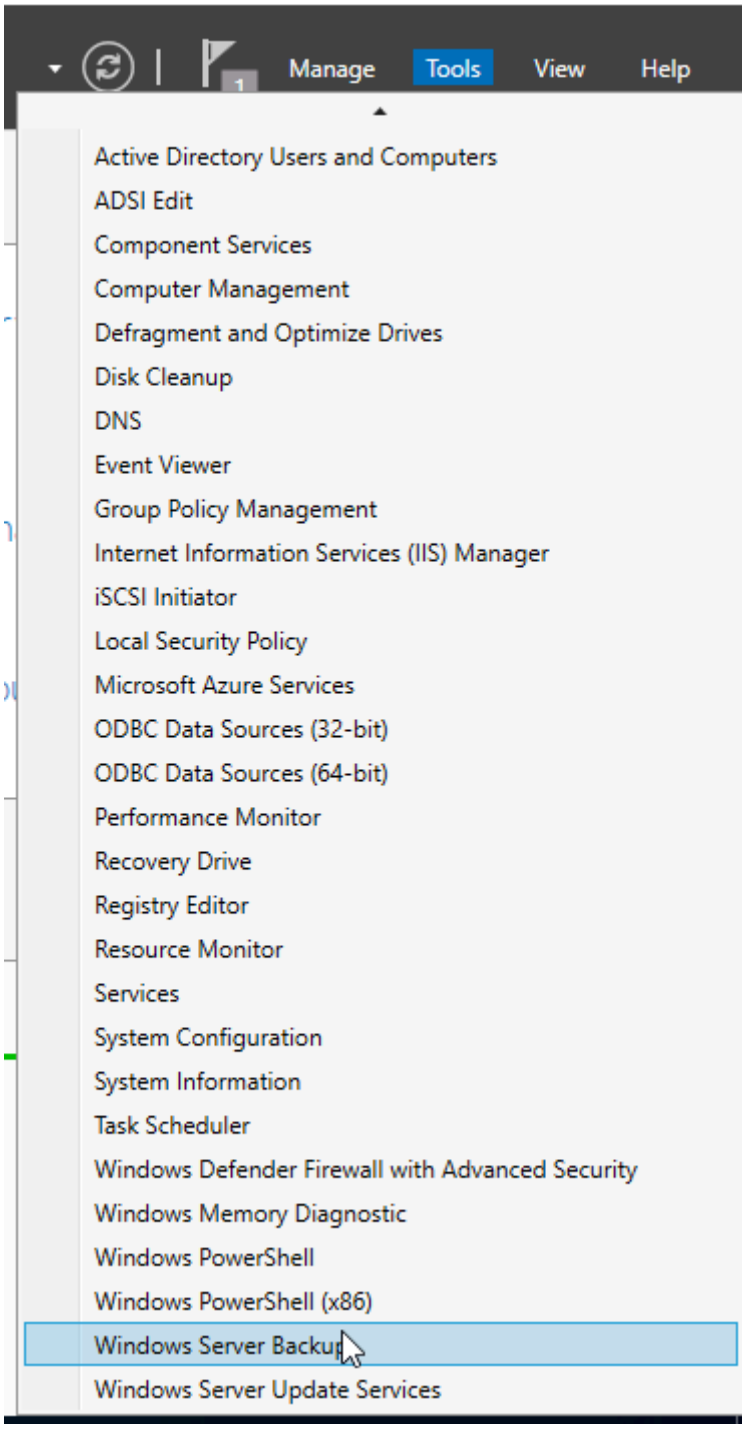
Instalar


Add Roles and Features Wizard
Confirm installation selections
DESTINATION SERVER
WSERVER57.midom57.et
Before You Begin
Installation Type
Server Selection
Server Roles
Features
Confirmation
Results
To install the following roles, role services, or features on selected server, click Install.
Restart the destination server automatically if required
Optional features (such as administration tools) might be displayed on this page because they have been selected automatically. If you do not want to install these optional features, click Previous to clear their check boxes.
Windows Server Backup
Export configuration settings
Specify an alternate source path
< Previous
Next >
Install
Cancel

📦 PASO 4 — Configurar copia automática programada

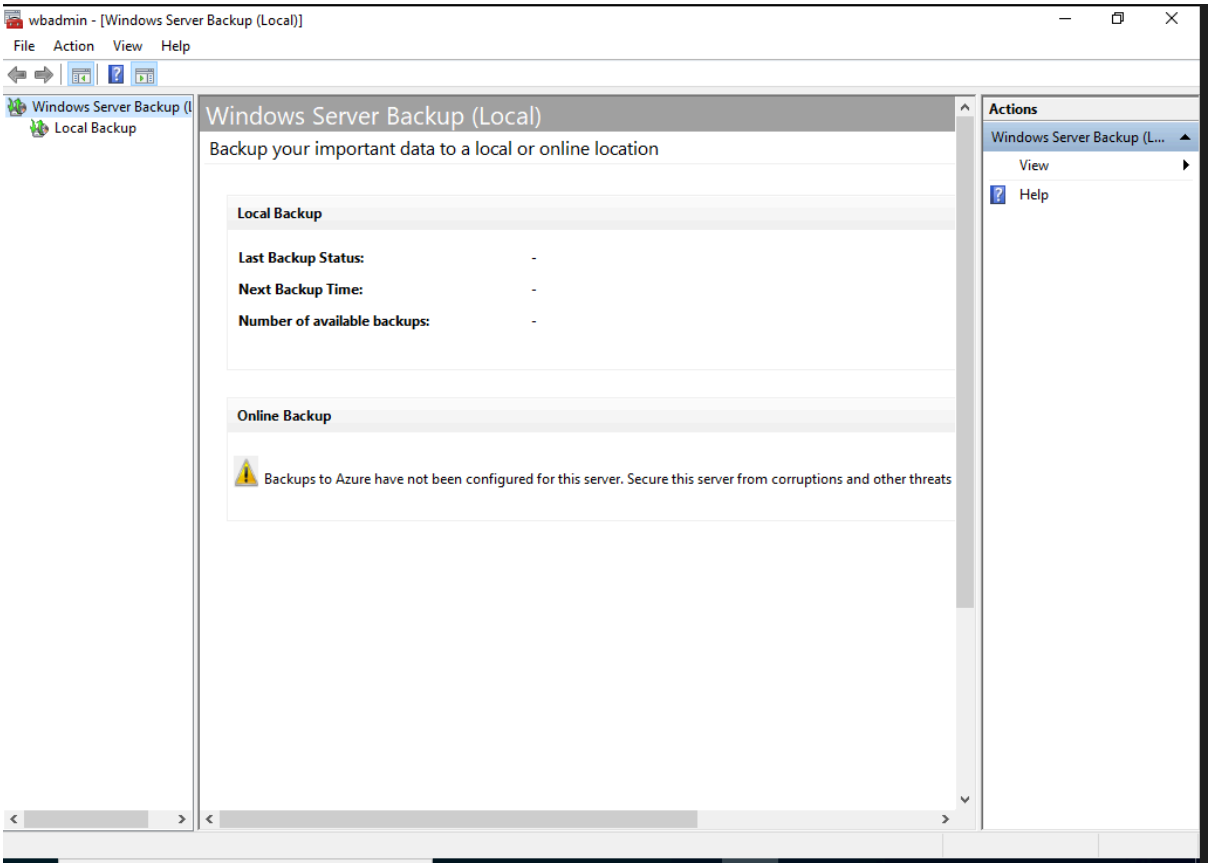
Abre:

Herramientas → Windows Server Backup

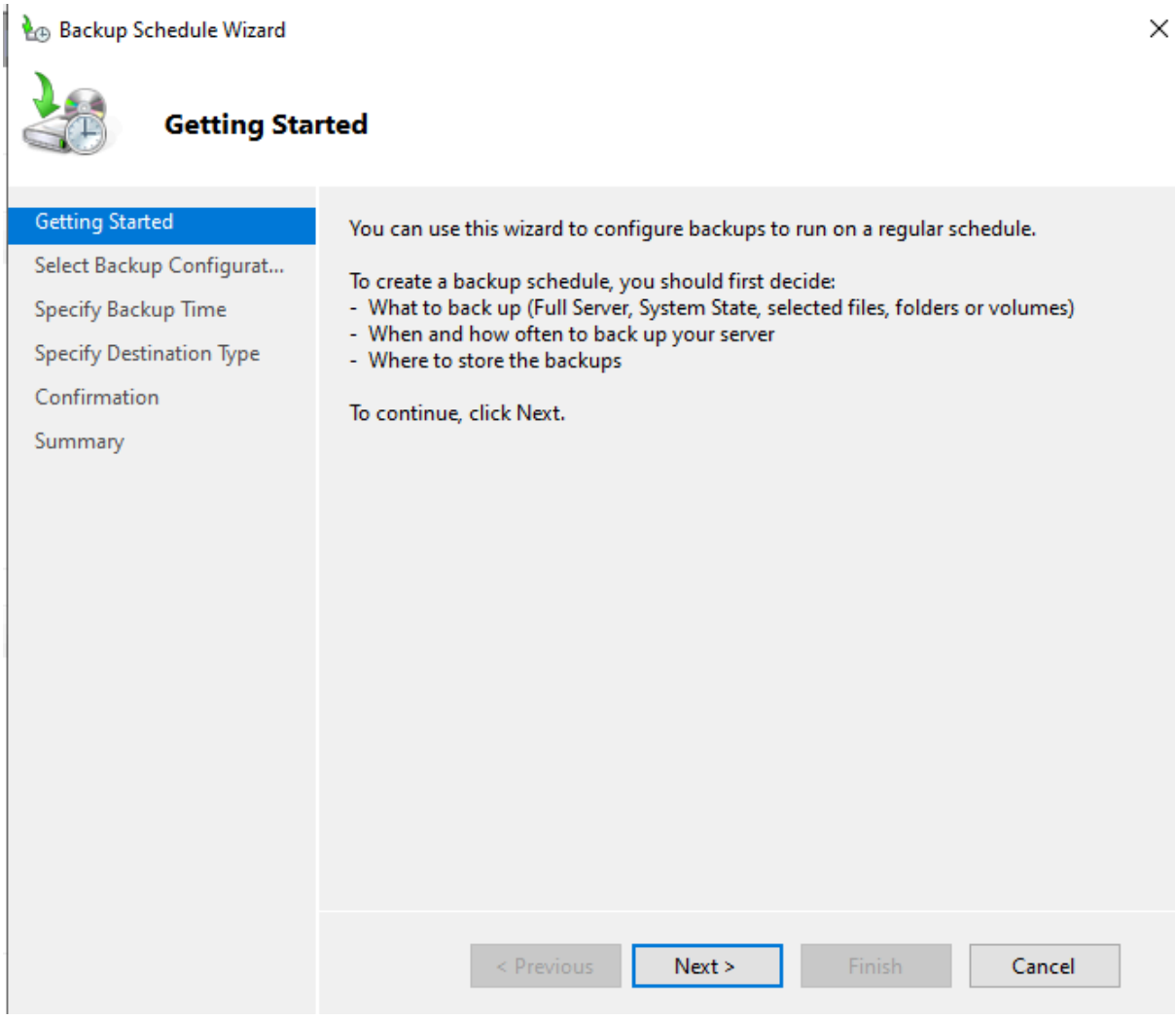


En el panel derecho:

Programación de copia de seguridad…



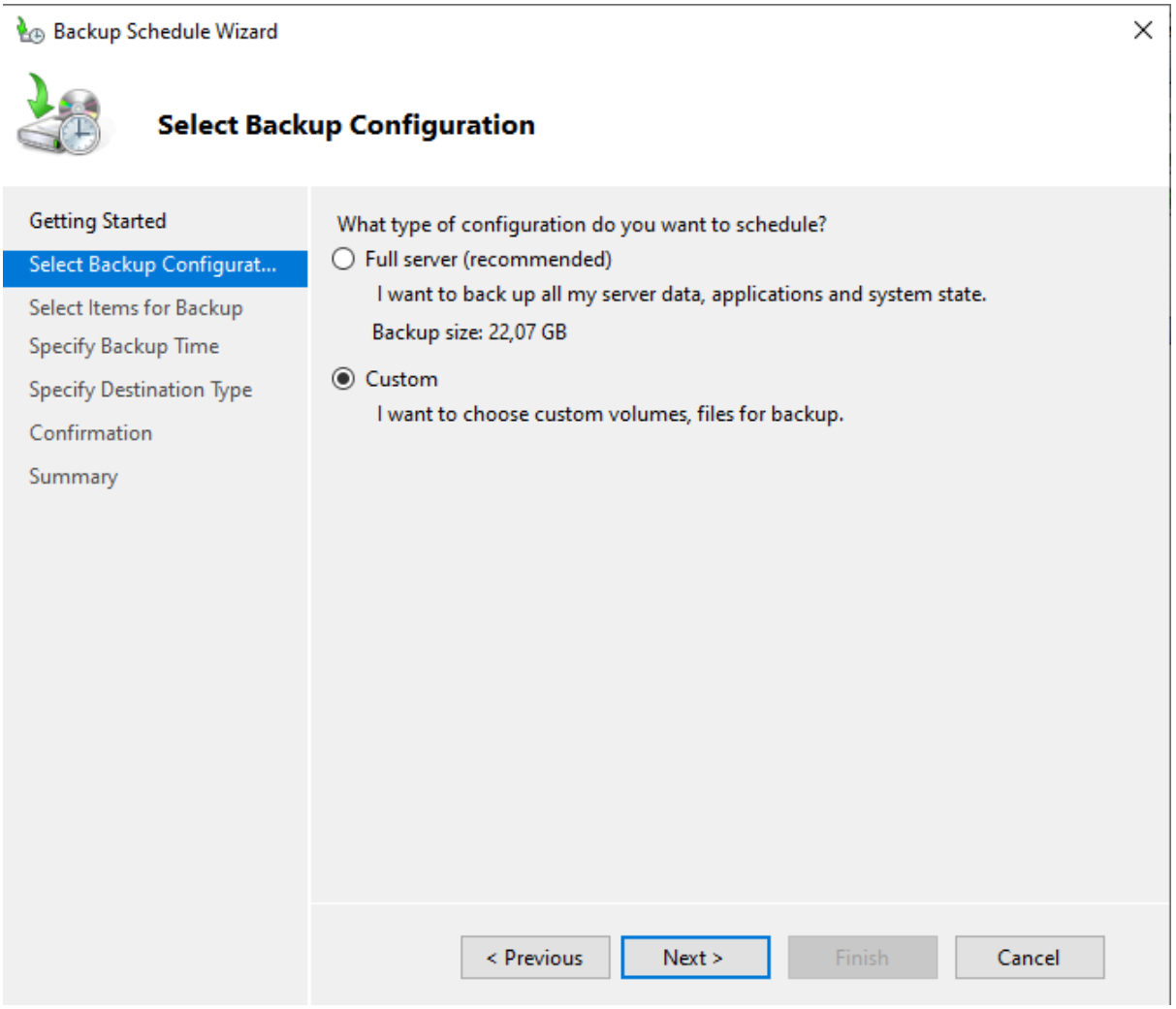
🧩 Asistente de configuración



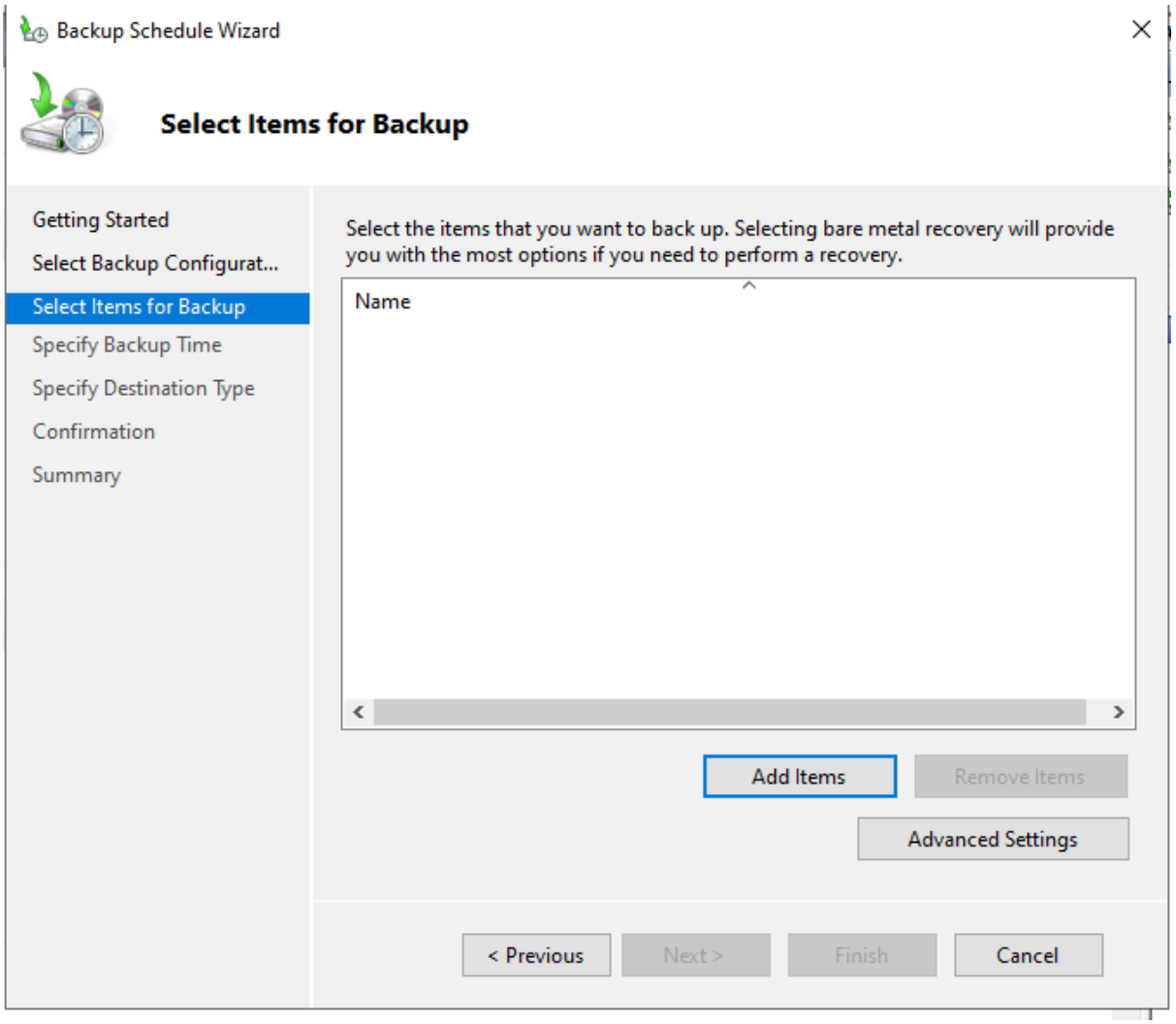
Tipo de configuración

Selecciona:

Personalizada

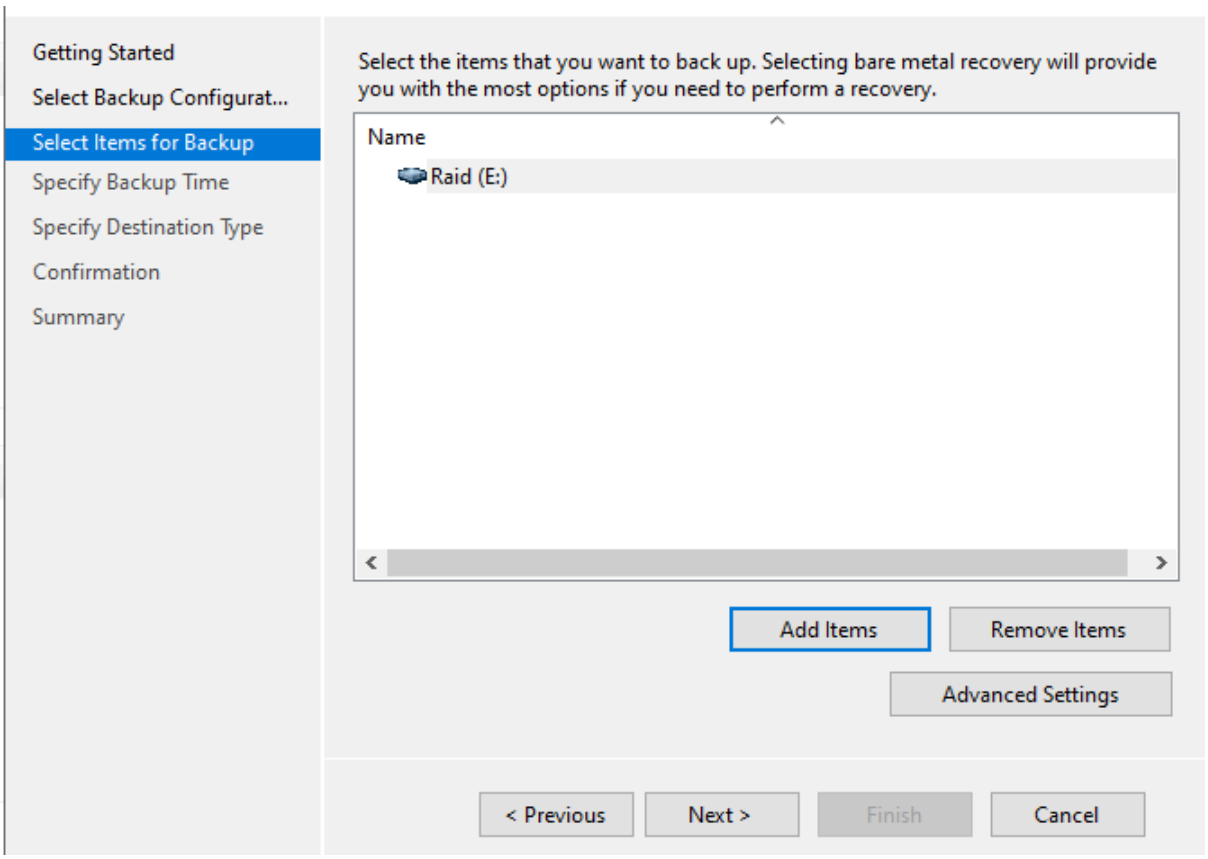

Backup Schedule Wizard
Select Backup Configuration
Getting Started
Select Backup Configurat...
Select Items for Backup
Specify Backup Time
Specify Destination Type
Confirmation
Summary
What type of configuration do you want to schedule?
Full server (recommended)
I want to back up all my server data, applications and system state.
Backup size: 22,07 GB
Custom
I want to choose custom volumes, files for backup.
< Previous
Next >
Finish
Cancel

Qué quieres respaldar



Marca:

- 📁 Carpetas compartidas (ej: `E:\Raid`)
- 📂 Volumen donde estén los recursos



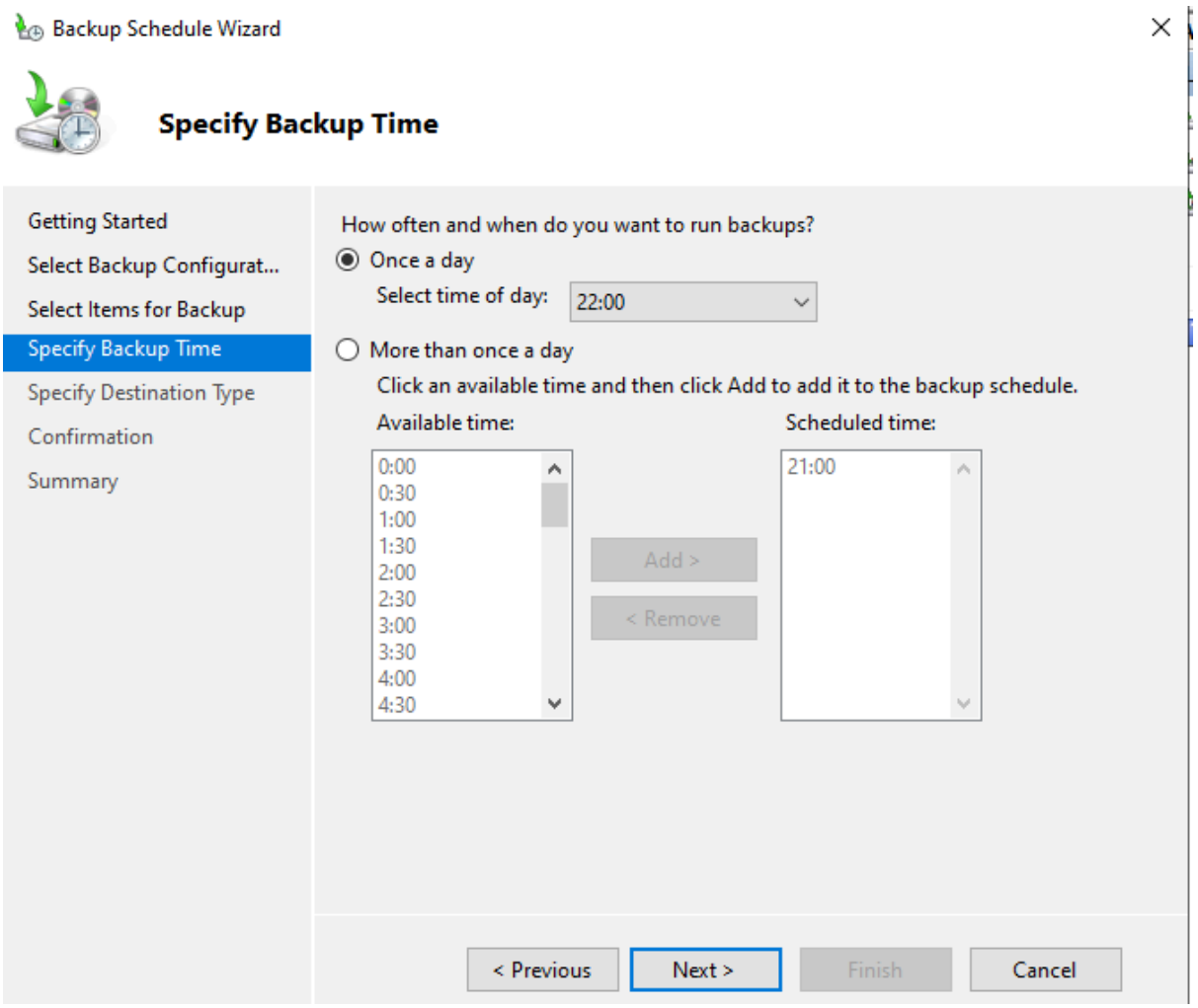
Si es servidor importante:

Puedes marcar **Estado del sistema**

Frecuencia

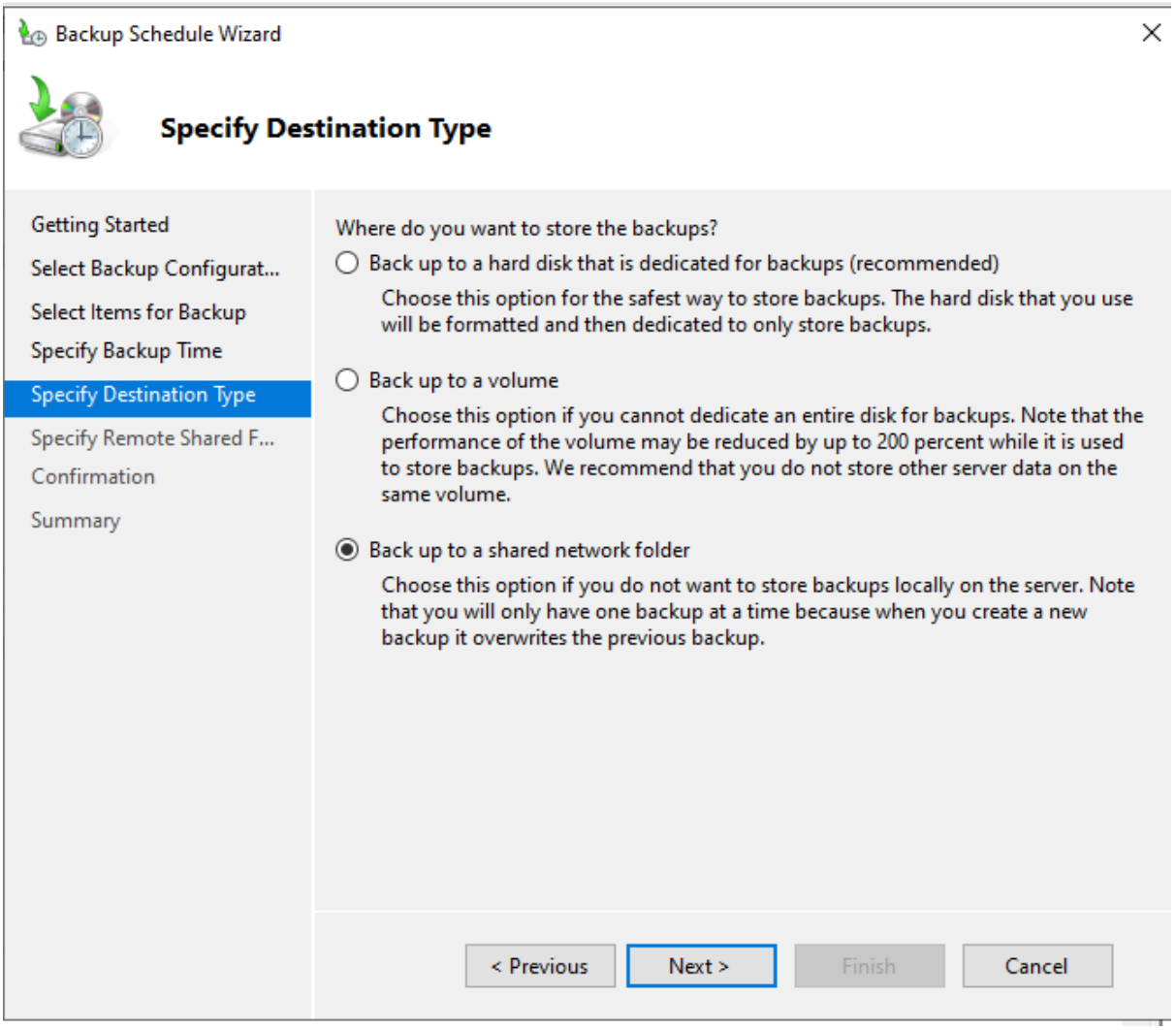
Te recomiendo:

- Una vez al día (por ejemplo 22:00)

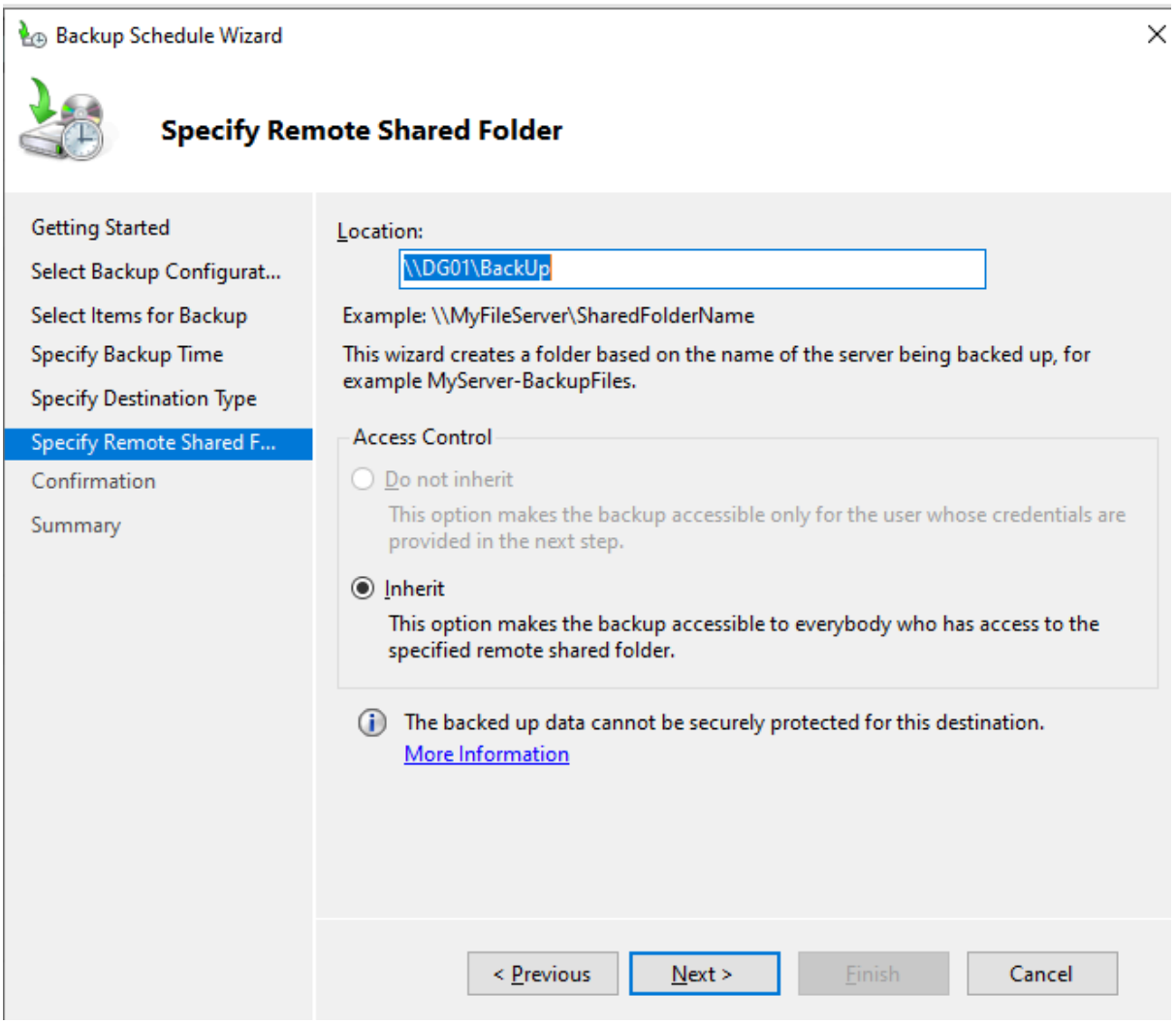


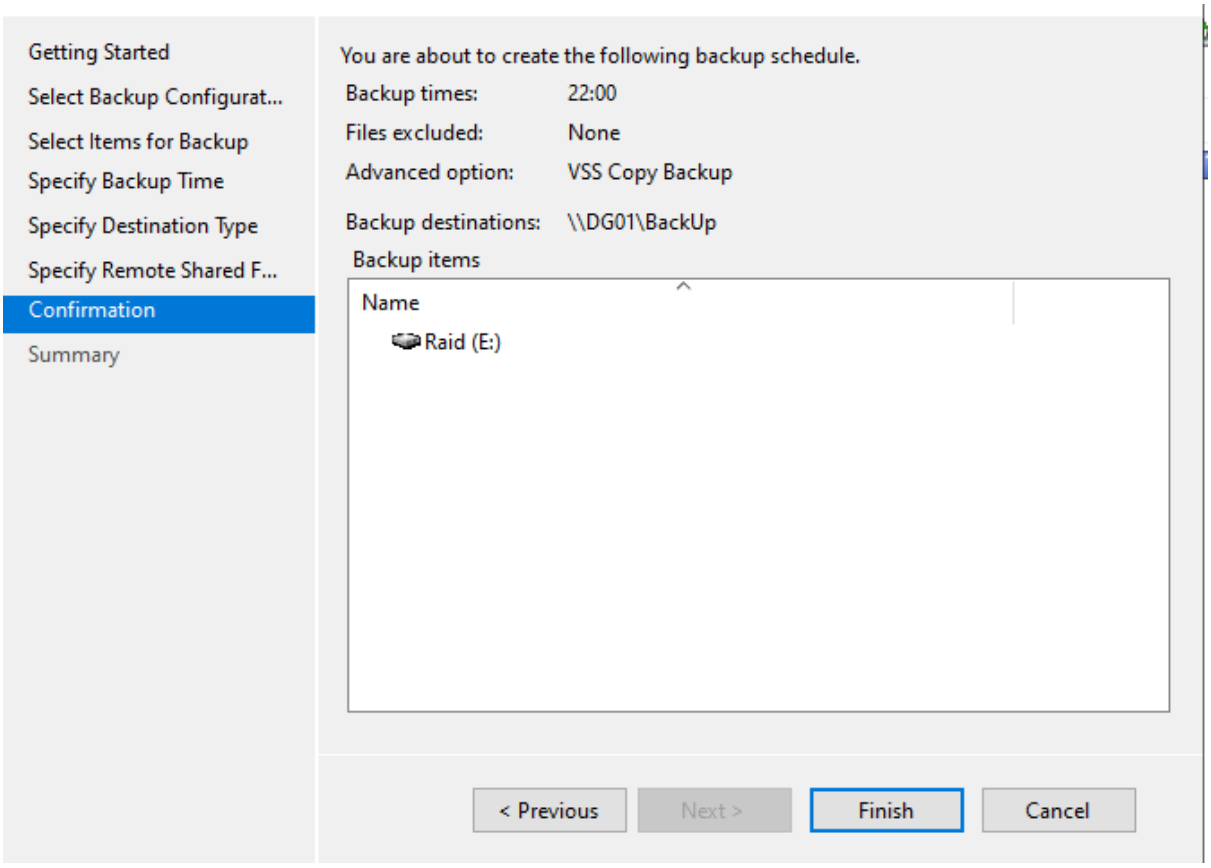
Destino del backup

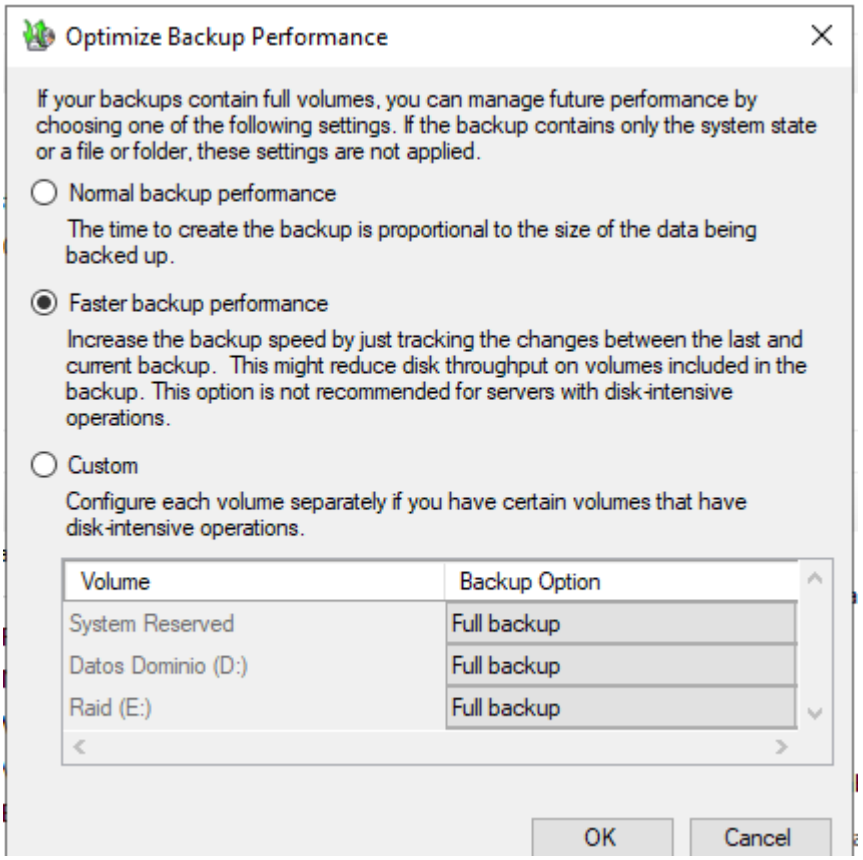
Selecciona:

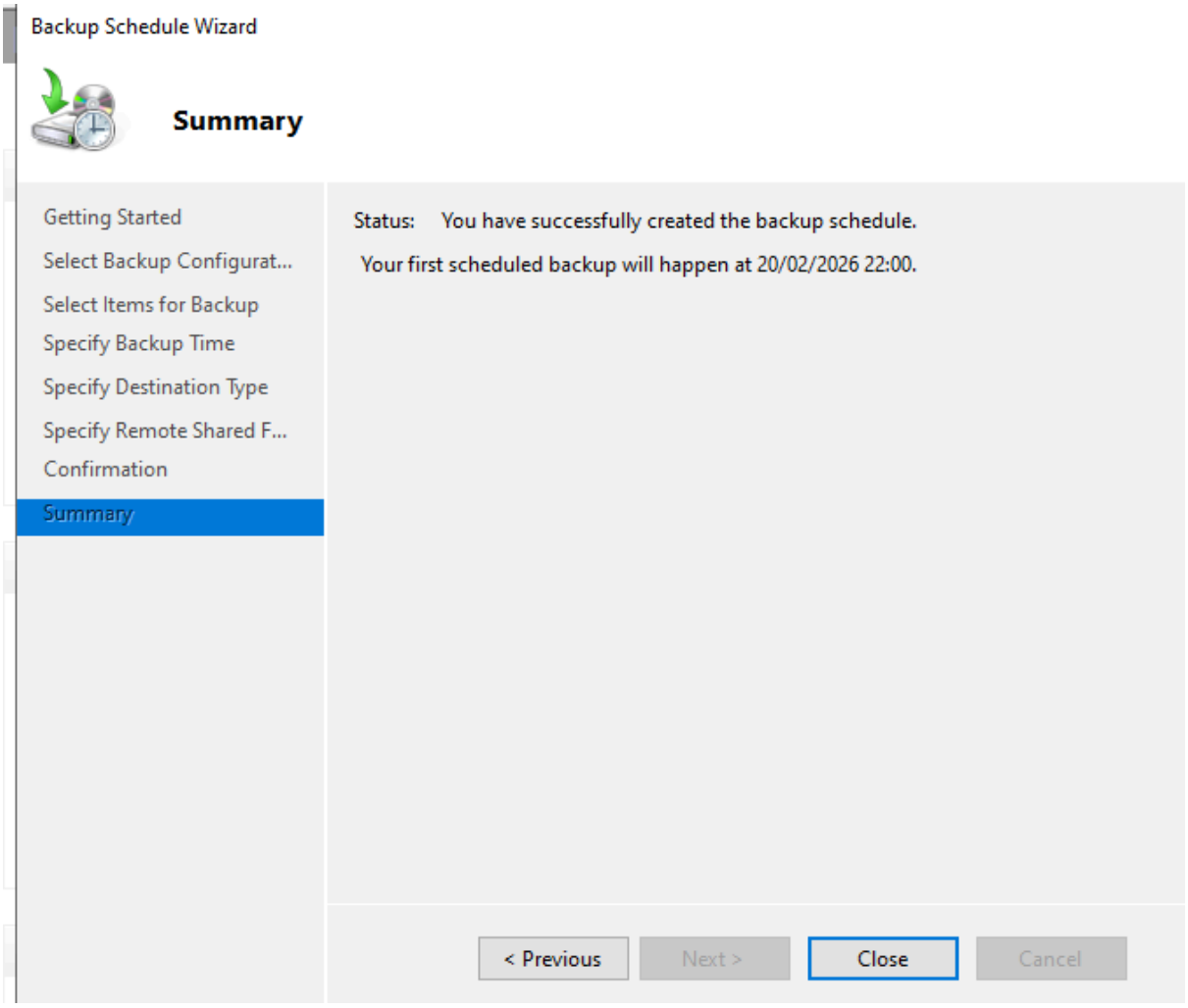


A continuación nos dirá donde queremos almacenar los BackUps, aquí es donde actuará la carpeta que hemos compartido anteriormente en la red.

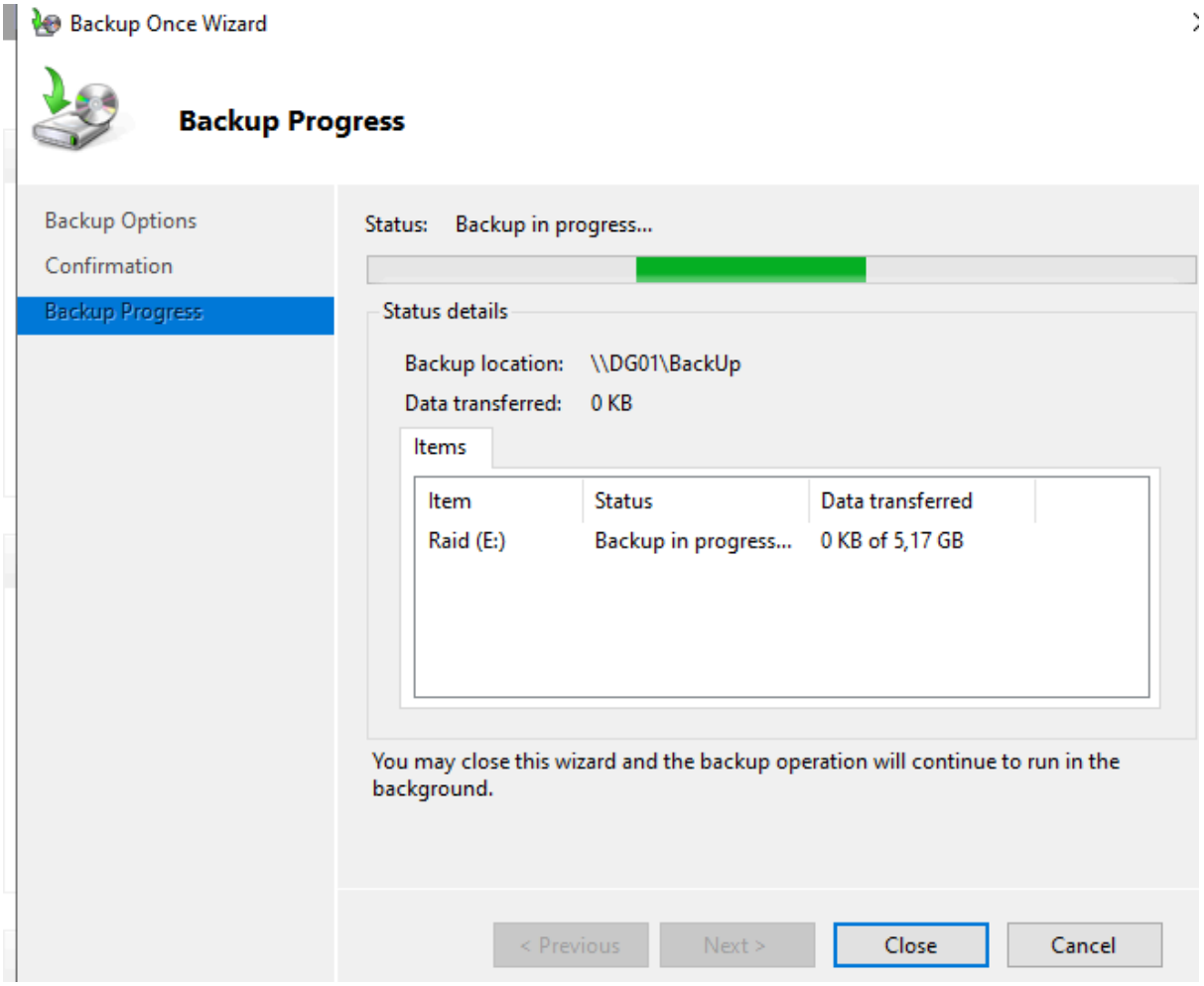

Backup Schedule Wizard
Specify Remote Shared Folder
Getting Started
Select Backup Configurat...
Select Items for Backup
Specify Backup Time
Specify Destination Type
Specify Remote Shared F...
Confirmation
Summary
Location:
\\DG01\BackUp
Example: \\MyFileServer\SharedFolderName
This wizard creates a folder based on the name of the server being backed up, for example MyServer-BackupFiles.
Access Control
Do not inherit
This option makes the backup accessible only for the user whose credentials are provided in the next step.
Inherit
This option makes the backup accessible to everybody who has access to the specified remote shared folder.
The backed up data cannot be securely protected for this destination.
More Information
< Previous
Next >
Finish
Cancel


Getting Started
Select Backup Configurat...
Select Items for Backup
Specify Backup Time
Specify Destination Type
Specify Remote Shared F...
Confirmation
Summary
You are about to create the following backup schedule.
Backup times: 22:00
Files excluded: None
Advanced option: VSS Copy Backup
Backup destinations: \\DG01\BackUp
Backup items
Name
Raid (E:)
< Previous
Next >
Finish
Cancel

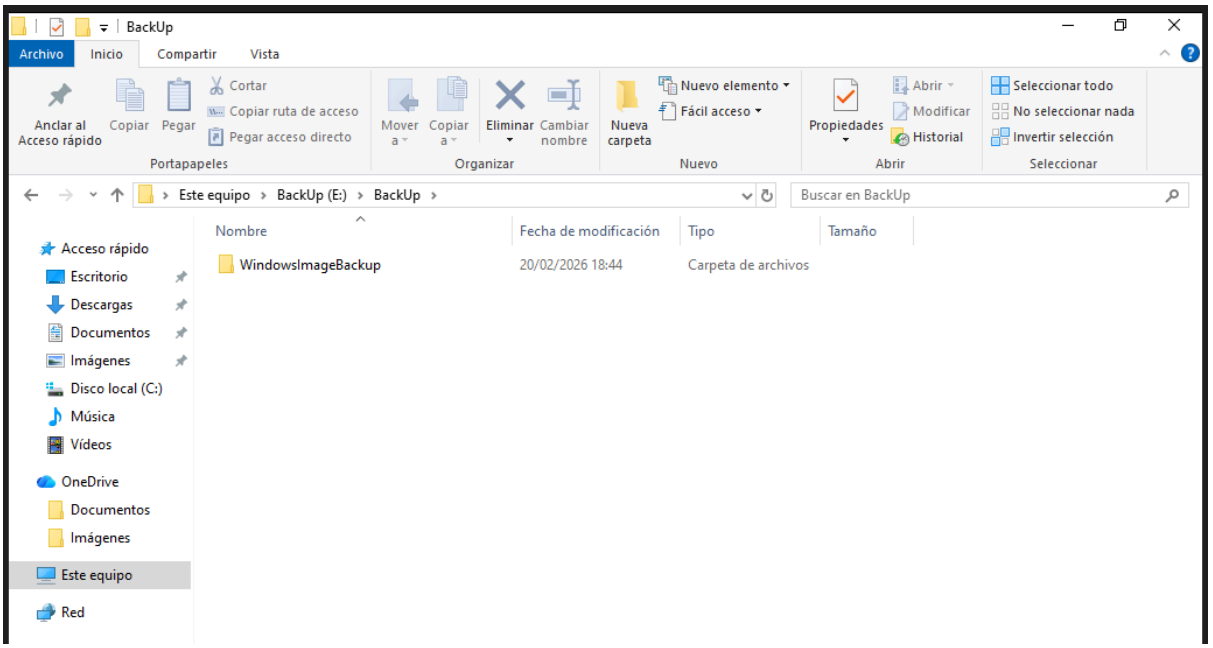

Optimize Backup Performance
If your backups contain full volumes, you can manage future performance by choosing one of the following settings. If the backup contains only the system state or a file or folder, these settings are not applied.
Normal backup performance
The time to create the backup is proportional to the size of the data being backed up.
Faster backup performance
Increase the backup speed by just tracking the changes between the last and current backup. This might reduce disk throughput on volumes included in the backup. This option is not recommended for servers with disk-intensive operations.
Custom
Configure each volume separately if you have certain volumes that have disk-intensive operations.
Volume
Backup Option
System Reserved
Full backup
Datos Dominio (D:)
Full backup
Raid (E:)
Full backup
OK
Cancel



Comprobación:



Vemos que a continuación nos lo crea en el cliente sin ningún tipo de problema:



🌐PASO 5 — Comando para copia completa

El comando base es:

```
wbadmin start backup -backupTarget:E: -include:D: -allCritical -vssFull -quiet
```

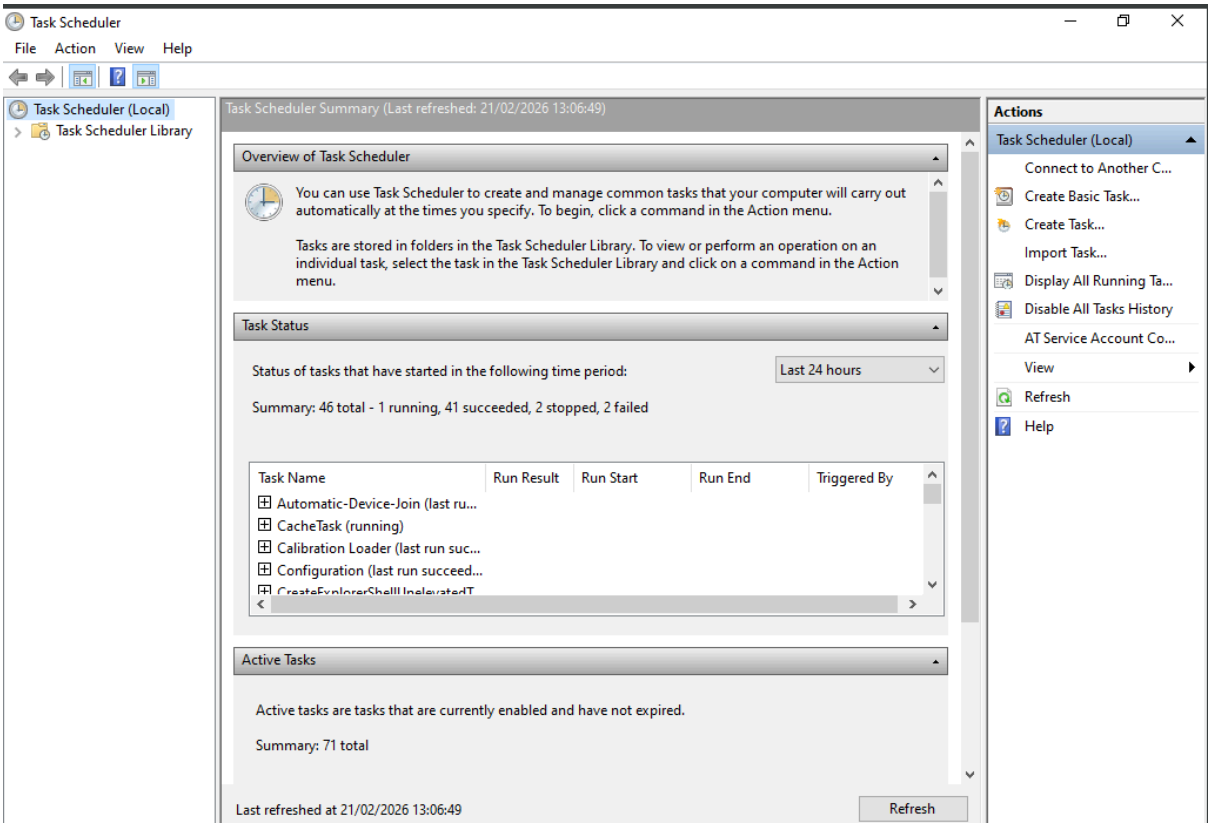
Explicación de qué hace cada cosa:

- `-backupTarget:E:` → Disco de backup (ajusta la letra)
- `-include:D:` → Volumen donde tienes los datos
- `-allCritical` → Incluye volúmenes críticos del sistema
- `-vssFull` → Hace copia completa VSS (marca los archivos como respaldados)
- `-quiet` → No pide confirmación

📅 PASO 6 — Crear tarea programada

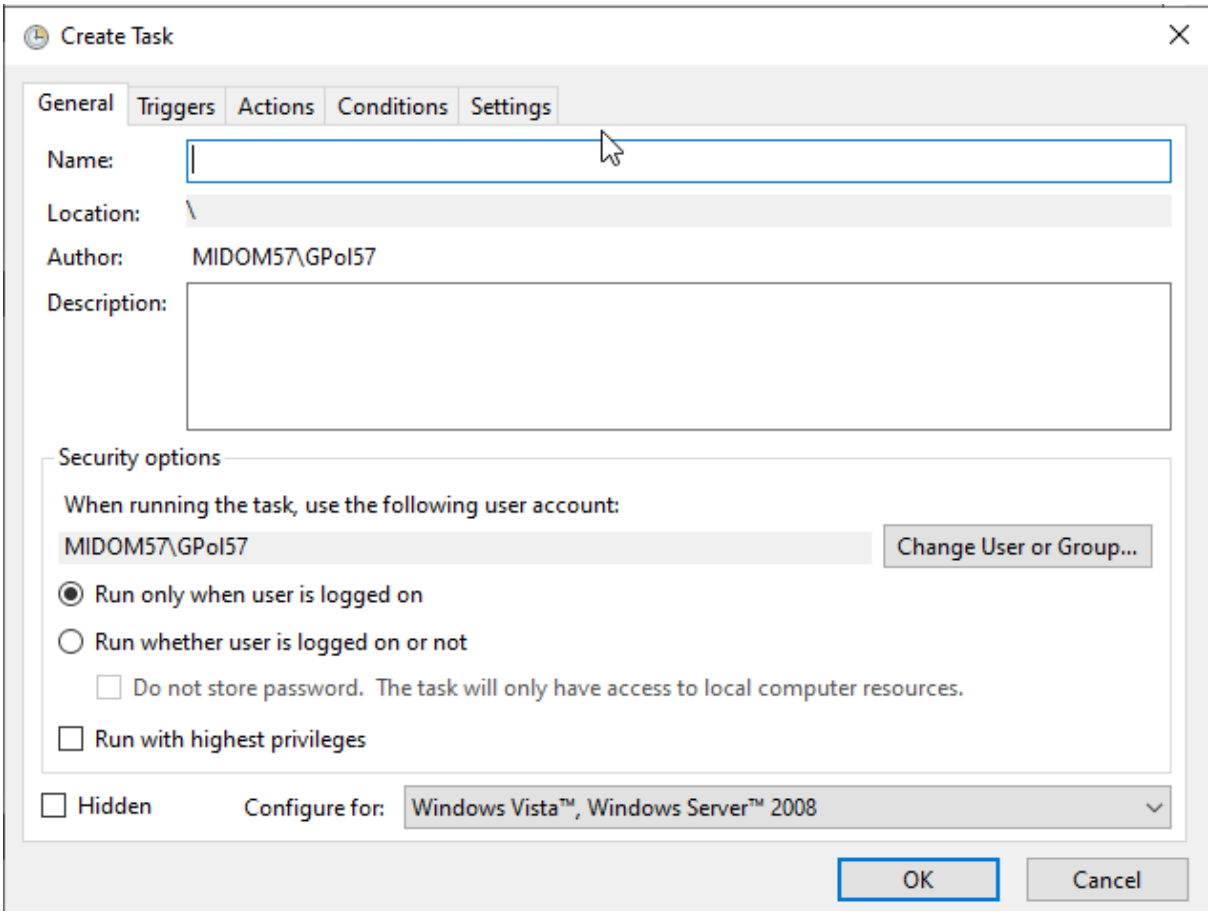
1. Abre:

   Programador de tareas



2. Haz clic en:

   Crear tarea…



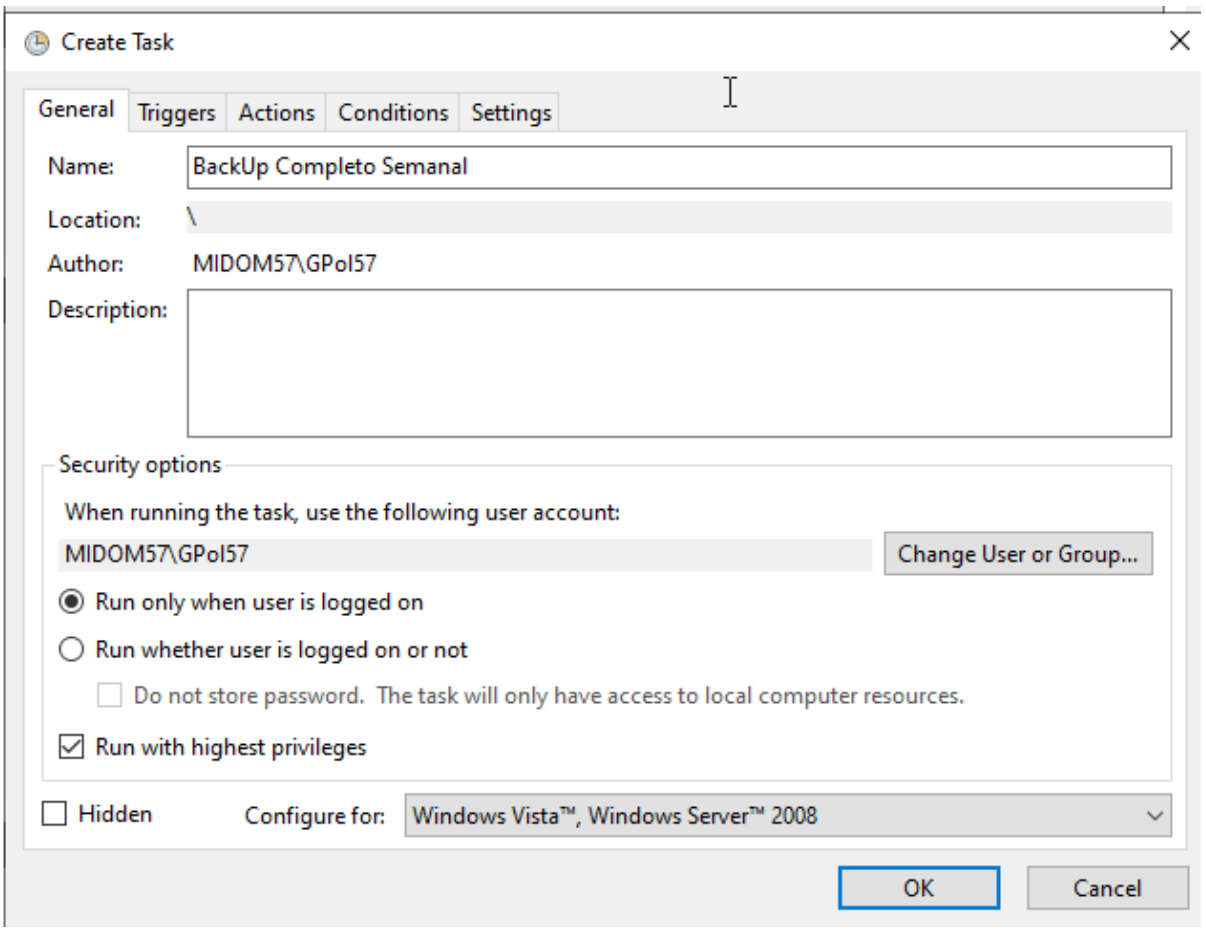
- Pestaña General

  - Nombre:

    Backup Completo Semanal

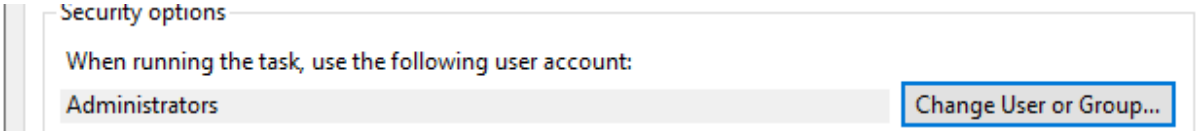
  - Marca:

    - ✅ Ejecutar tanto si el usuario inició sesión como si no
    - ✅ Ejecutar con los privilegios más altos



- Usuario:

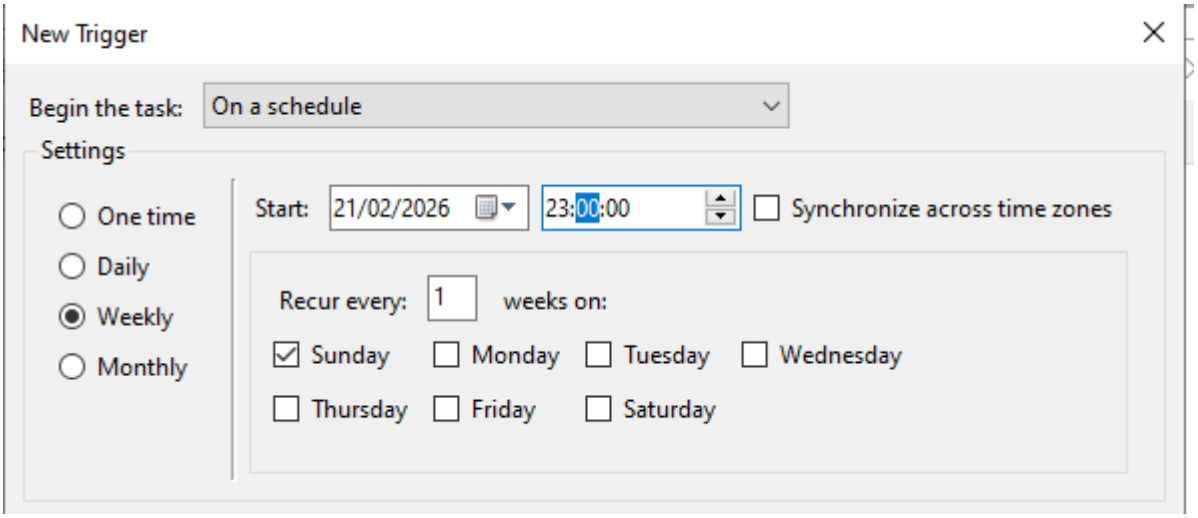
  → Administrator



- Pestaña Desencadenadores

1. Nuevo…
2. Configura:
   - Semanal
   - Domingo
   - 23:00 (por ejemplo)

Aceptar.

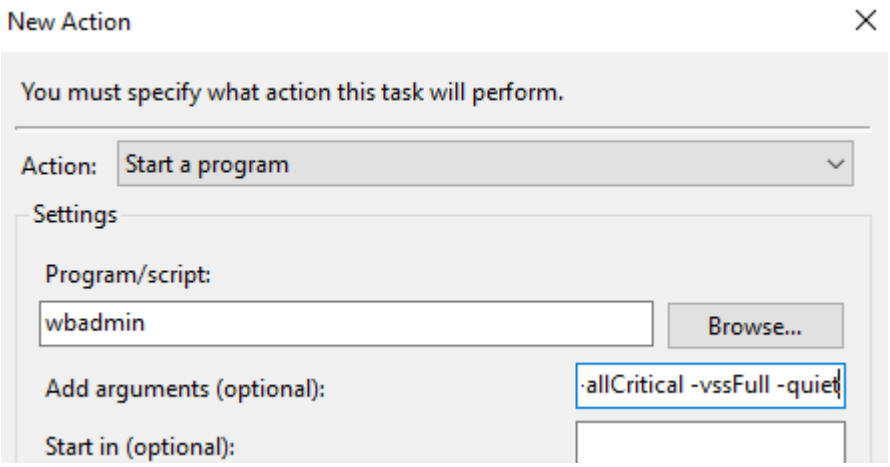


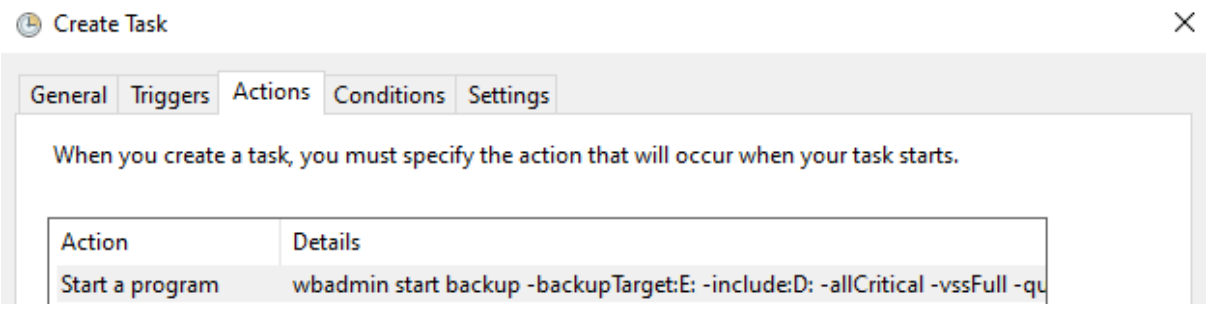
- Pestaña Acciones

1. Nueva…
2. Acción:

→ Iniciar un programa

Programa o script:

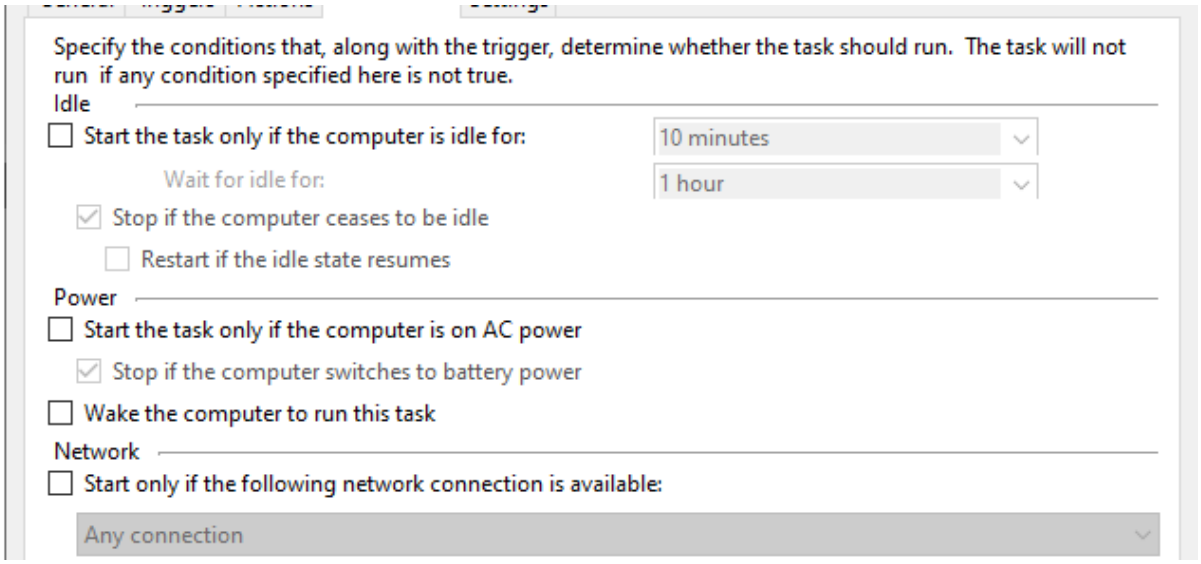




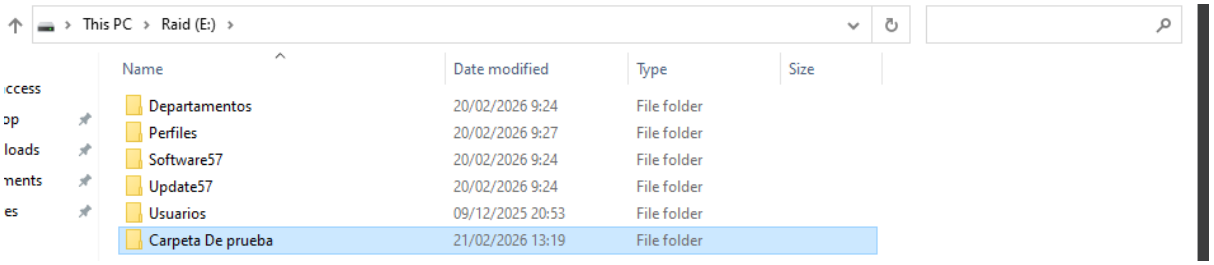
🔹 Pestaña Condiciones

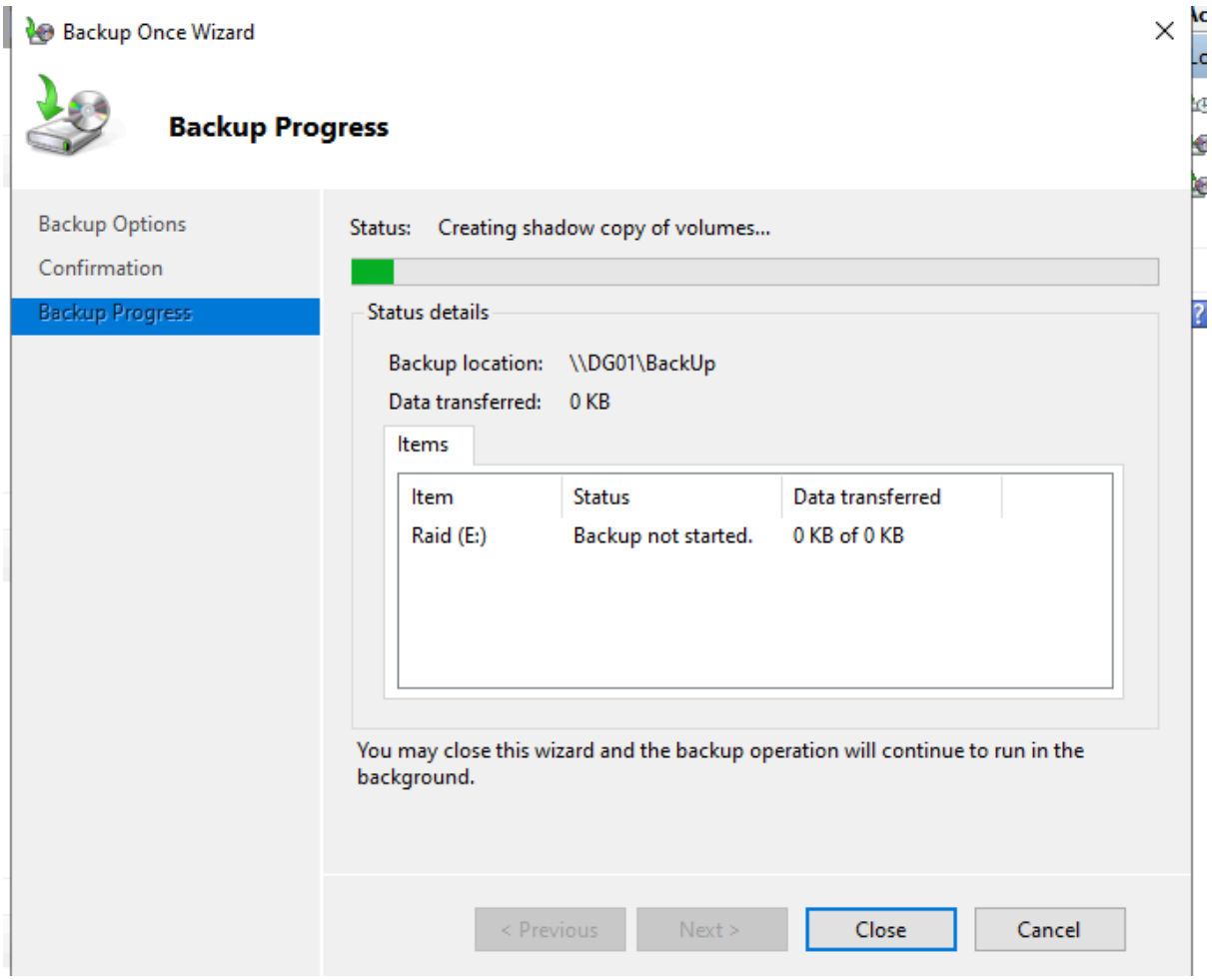
Opcional:

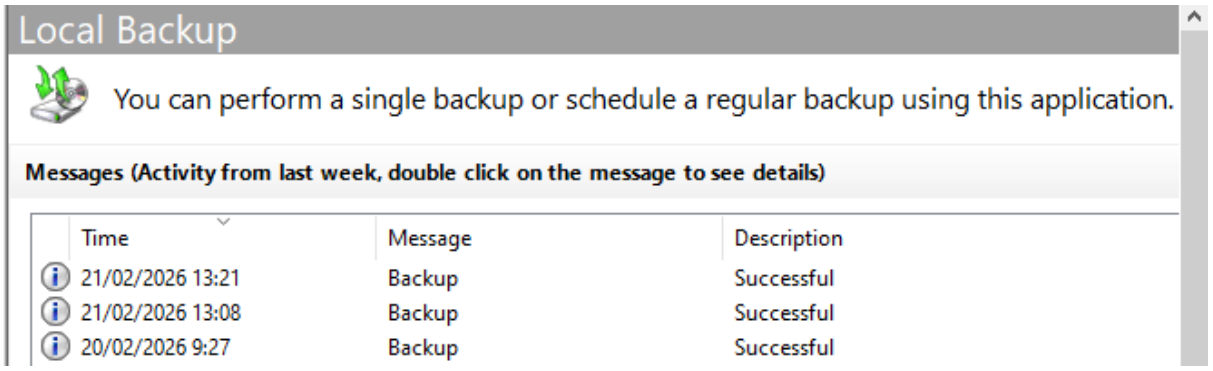
- Quita lo de “solo si está conectado a corriente” si es servidor virtual.



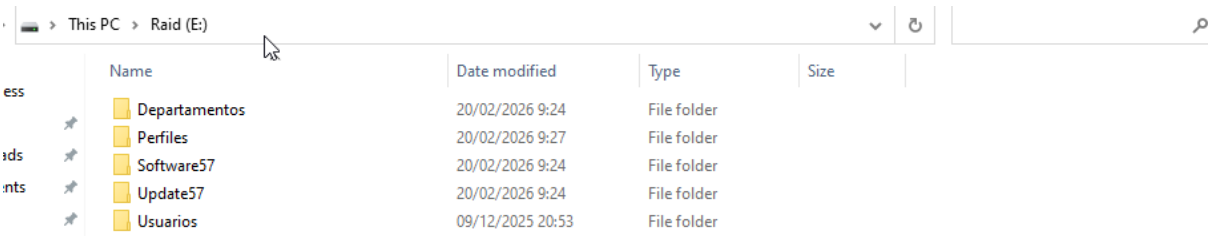
✅ PASO 7 — Verificar que funciona





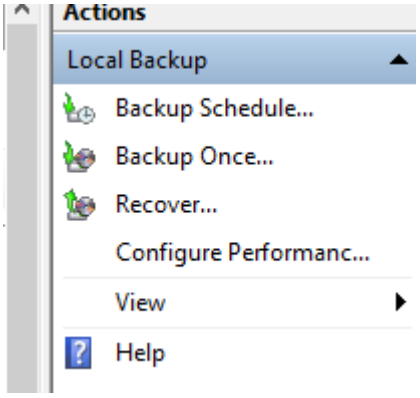


Eliminamos la carpeta a continuación:



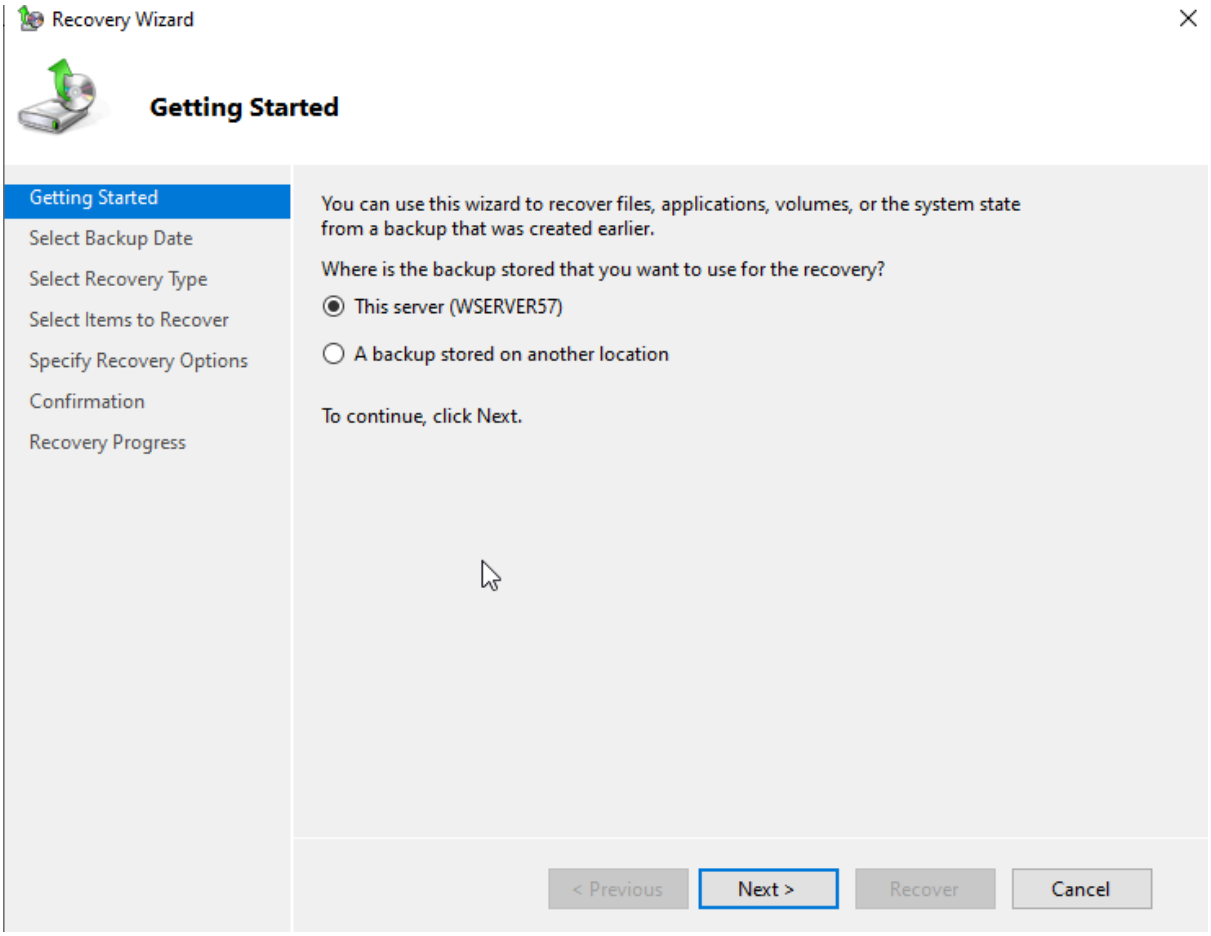
🔄 PASO 8 — Comprobar restauración (MUY IMPORTANTE)

1. Ve a:

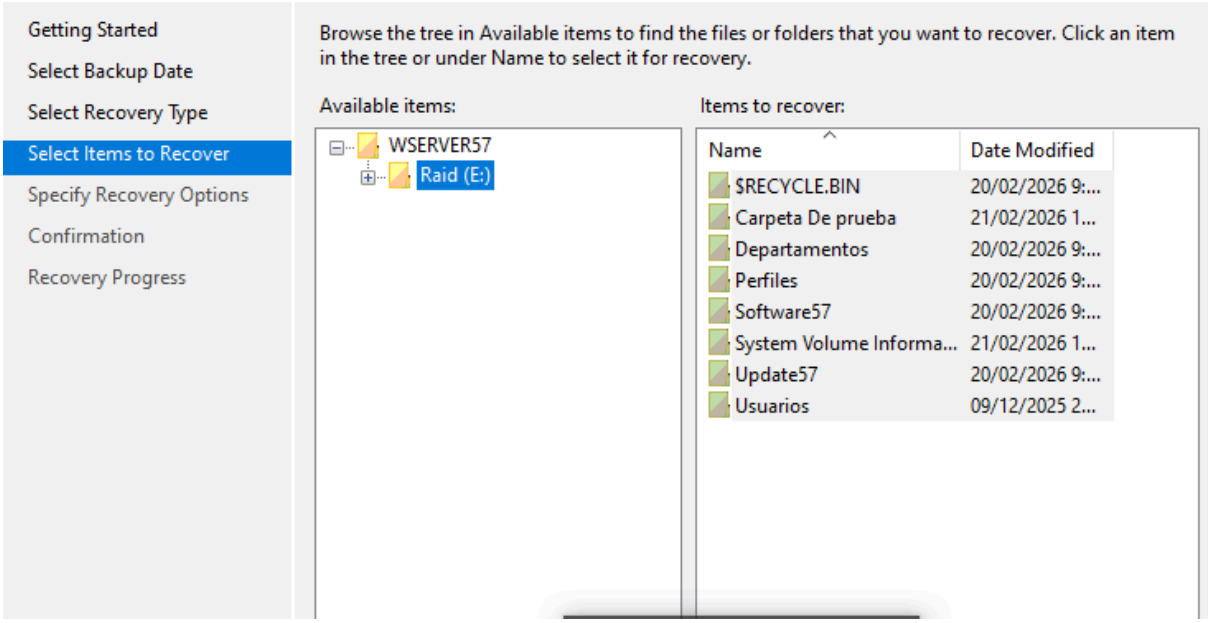


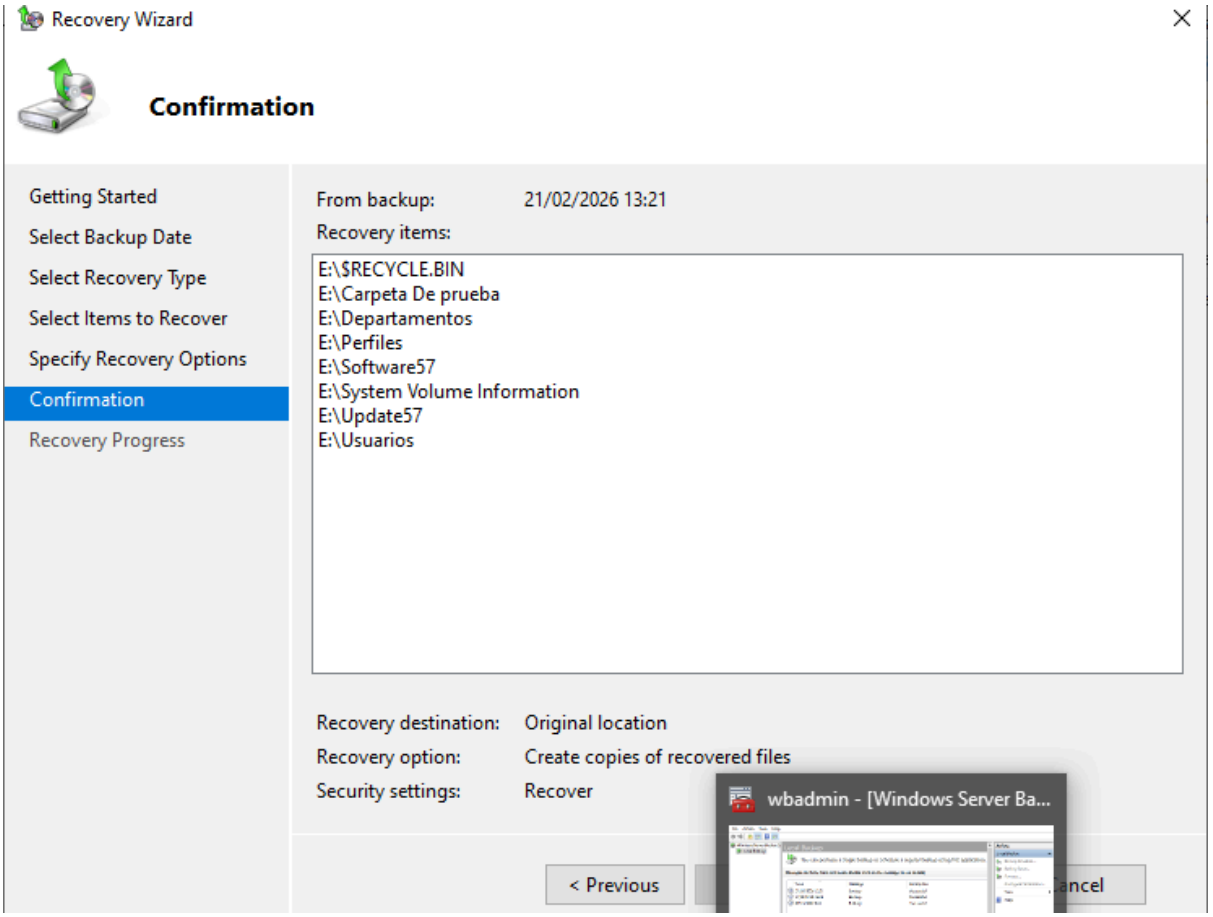
Selecciona:

- Este servidor
- Última copia

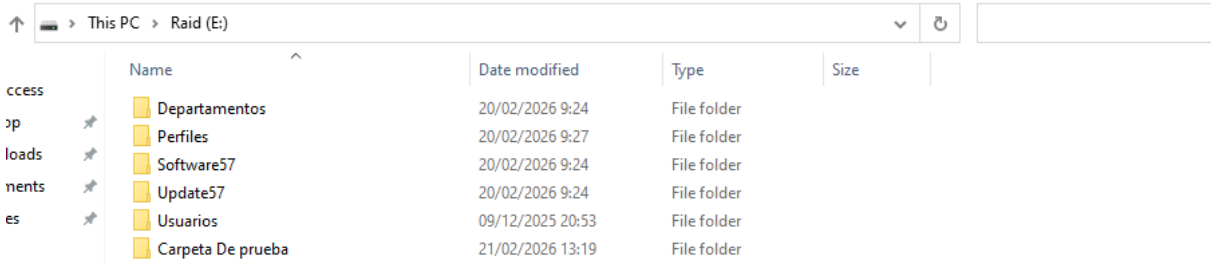


Restaura el archivo.



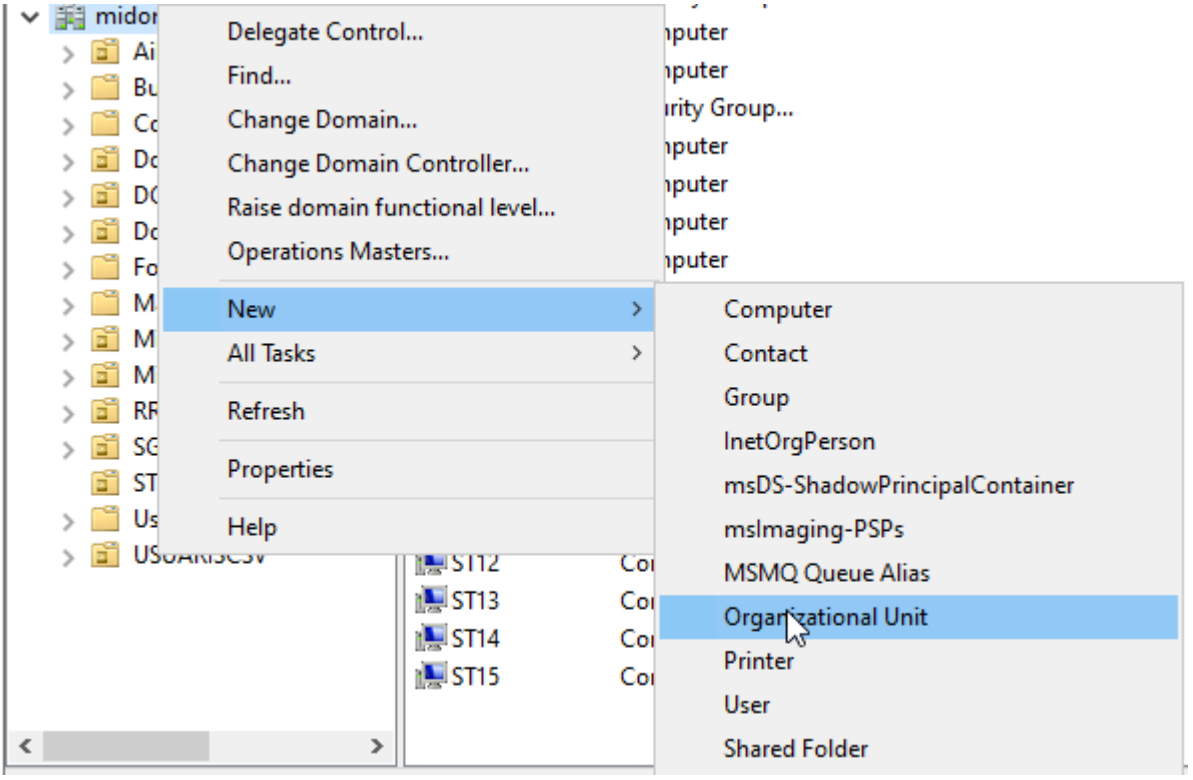


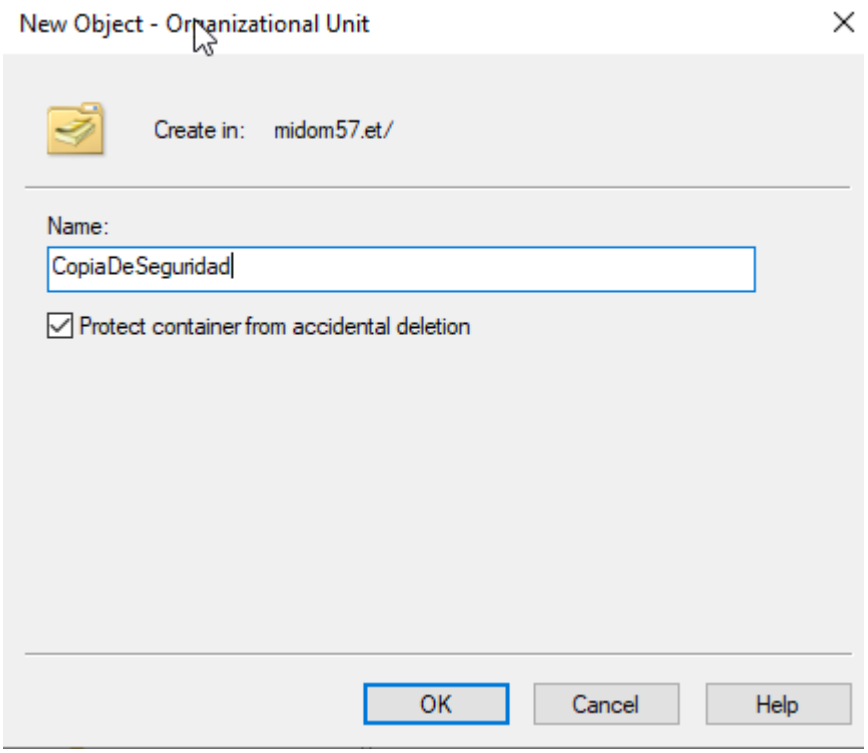
Una vez acabe de restaurar la copia de seguridad revisaremos y tendremos la carpeta:



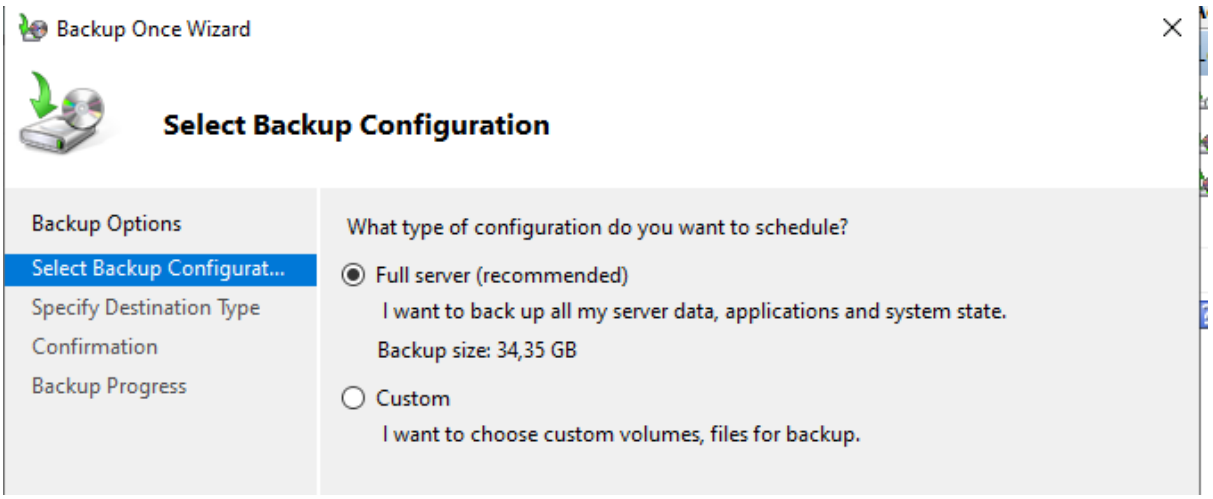
Caso Active Directory

Pongamos de que el caso borramos una Unidad Organizativa por accidente y la queremos restaurar mediante a una copia de seguridad a continuación veremos como se puede hacer:

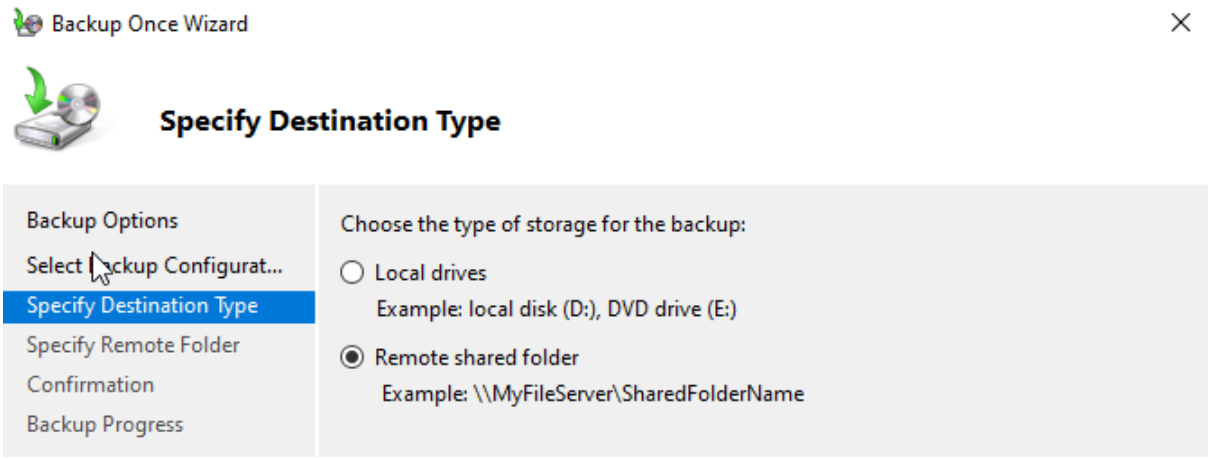


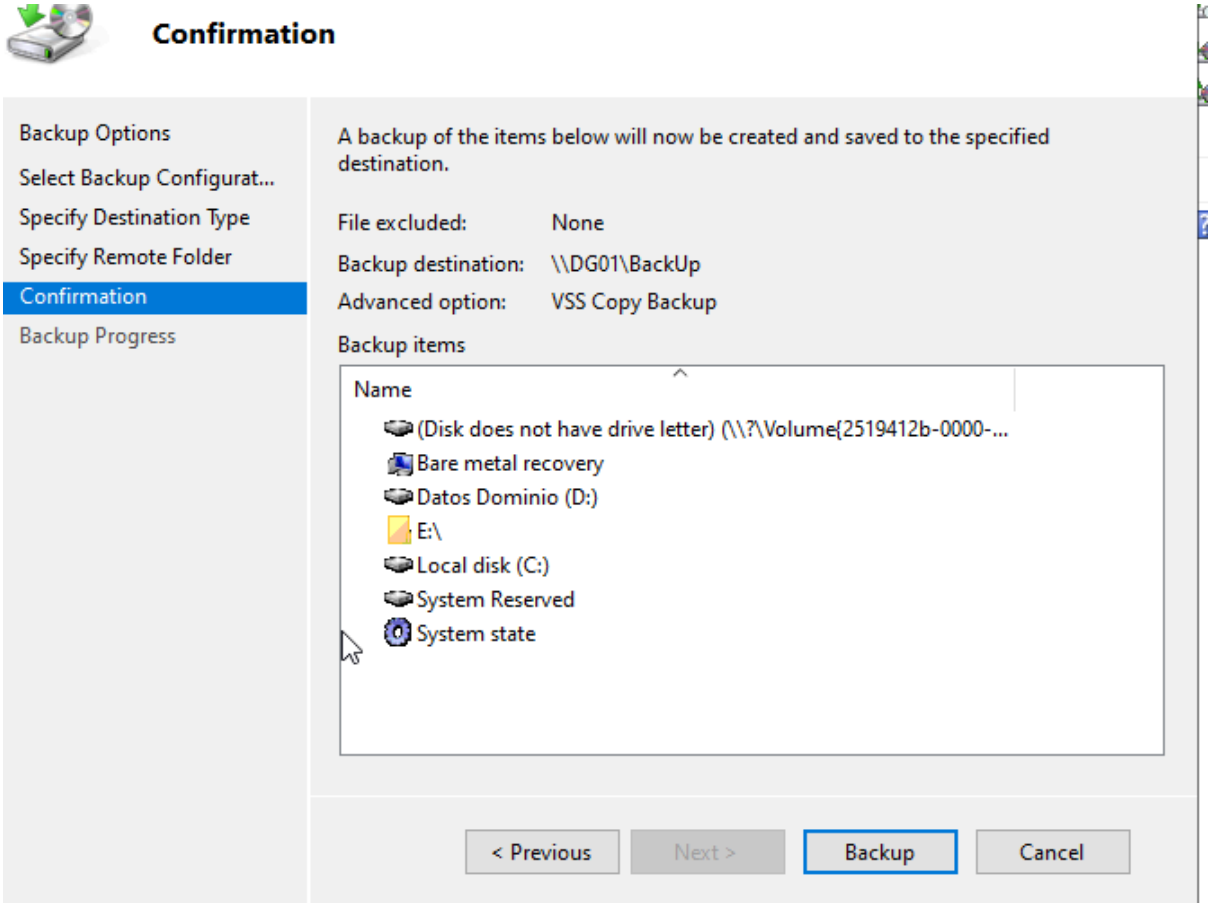


Una vez tengamos la Unidad Organizativa haremos una copia de seguridad total del disco C:\ donde se encuentran los datos del AD

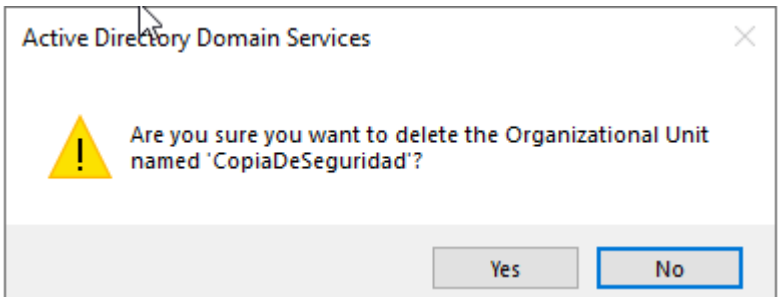


Mandaremos la copia de seguridad a la carpeta compartida que tenemos.

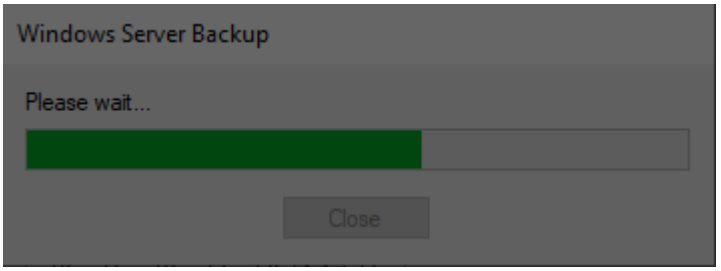
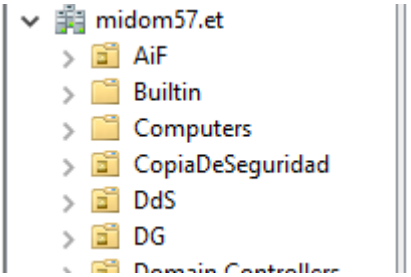




Una vez acabe la copia de seguridad completa del servidor probaremos a borrar la unidad organizativa y a restaurar la copia a ver que pasa.



Una vez hecho deberemos de restaurar la copia de seguridad desde el backup.



Una vez restaurada entraremos en el Panel de gestión de elementos de AD y veremos si ha vuelto a aparecer la unidad organizativa.